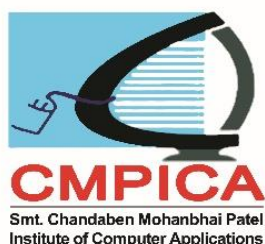




ACADEMIC REGULATIONS & SYLLABUS

Faculty of Computer Science & Applications
Smt. Chandaben Mohanbhai Patel Institute of
Computer Applications
M.Sc.(IT) Programme





CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Computer Science & Applications

ACADEMIC REGULATIONS

M.Sc. (IT) Programme
(Choice Based Credit System)

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in
www.charusat.ac.in

Year – 2021-2022

CHARUSAT

FACULTY OF COMPUTER SCIENCE AND APPLICATIONS

ACADEMIC RULES

Master of Science in Information Technology (M.Sc. (IT).) Programme

To ensure uniform system of education, duration of post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working days duration. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

2. Duration of Programme

Postgraduate programme : Master of Science in Information Technology

Minimum	4 semesters (2 academic years)
Maximum	6 semesters (3 academic years)

3. Eligibility & Mode of admissions

Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Government of Gujarat from time to time.

4. Programme structure and Credits

A student admitted to a program should study the course and earn credits specified in the course structure.

5. Attendance

- 5.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.
- 5.2 Student attendance in a course should be 80%.

6 Course Evaluation

- 6.1 The performance of every student in each course will be evaluated as follows:
- 6.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and
 - 6.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.
- 6.2 University Examination
- 6.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.
 - 6.2.2 In order to earn the credit in a course a student has to obtain grade other than FF.
- 6.3 Performance at Internal & University Examination will be done on the relative grading system.

7 Grading

The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are defined as follows:

$$\text{SGPA} = \frac{\sum C_i G_i}{\sum C_i} \quad \text{where } C_i \text{ is the number of credits of course } i \\ G_i \text{ is the Grade Point for the course } i \\ \text{and } i = 1 \text{ to } n, n = \text{number of courses in the semester}$$

$$\text{CGPA} = \frac{\sum C_i G_i}{\sum C_i} \quad \text{where } C_i \text{ is the number of credits of course } i \\ G_i \text{ is the Grade Point for the course } i$$

and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

8 Detention Rule

8.1 No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.

8.2 A Student will not be allowed to move to third year if he/she has not cleared all the courses of first year.

8.3 A student will not be allowed to move to fourth year if he/she has not cleared all the courses of first and second year.

9 Awards of Degree

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and

9.1.2 He should have cleared all evaluation components in every course; and

9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for the award of degree. Only latest grade will be considered.

10 Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	$CGPA \geq 7.5 \ \& \ \leq 10$
First class:	$CGPA \geq 6.0 \ \& \ < 7.5$
Second Class:	$CGPA \geq 5.0 \ \& \ < 6.0$
Pass Class:	$CGPA < 5.0$

II Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHOICE BASED CREDIT SYSTEM
FOR
MASTER OF SCIENCE (INFORMATION TECHNOLOGY)

The choice based credit system provides flexibility in designing curriculum and assigning credits based on the course content and hour of teaching. The choice based credit system provides an opportunity for the students to choose courses from the prescribed courses comprising core, elective and open elective courses. The CBCS provides a cafeteria type approach in which the students can take courses of their choice and adopt an interdisciplinary approach to learning. The courses shall be evaluated on the grading system, which is considered to be better than the conventional marks system.

CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 3 types of courses:

Foundation Courses, Core Courses and Elective Courses.

Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core Courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme.

B. Programme Core Courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes.

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization.

C. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

Vision:

To become a leading institution in the field of computer applications and contribute in national efforts of computerizing public systems

Mission

To produce competent computer professionals with the ability to face future challenges.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)- MCA and M.Sc.(IT) Program

The graduates will

PEO1: Be able to understand and analyse the requirements of complex real world problems and solve them by designing optimised computer applications.

PEO2: Be able to apply skills in order to excel at professional careers.

PEO3: Be able adopt at rapidly changing technological landscape in domains of their preference.

PEO4: Be able to act ethically and professionally while maintaining high moral values.

PROGRAM OUTCOMES (POs)- MCA and M.Sc.(IT) Program

The Graduate of Computer Science and Applications will be able to :

PO1. Computational Knowledge: Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.

PO2. Problem Analysis: Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.

PO3. Design / Development of Solutions: Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand and propose integrated solutions using emerging technologies

PO4. Conduct Investigations of Complex Computing Problems: Ability to devise and conduct experiments, interpret data and provide well informed conclusions.

PO5. Modern Tool Usage: Ability to select modern computing tools, skills and techniques necessary for innovative software solutions

PO6. Professional Ethics: Ability to apply and commit professional ethics and cyber regulations in a global economic environment.

PO7. Life-long Learning: Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.

PO8. Project Management and Finance: Ability to understand, management and computing principles with computing knowledge to manage projects in multidisciplinary environments.

PO9. Communication Efficacy: Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.

PO10. Societal & Environmental Concern: Ability to recognize economical, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.

PO11. Individual & Team Work: Ability to work as a member or leader in diverse teams in multidisciplinary environment.

PO12. Innovation and Entrepreneurship: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.

PROGRAM SPECIFIC OUTCOMES (PSOs)- MCA and M.Sc.(IT) Program

At the end of the programme, the student should be able to

PSO1: Analyse, design and develop effective software applications for solving contemporary challenges.

PSO2: Harness the necessary skills with spirit of research and entrepreneurship.

**TEACHING SCHEME
FOR
M.Sc. (IT) PROGRAMME
(1ST & 2ND YEAR)

EFFECTIVE FROM
ACADEMIC YEAR 2021-22**

Smt. Chandaben Mohanbhai Patel Institute of Computer Applications

Teaching and Examination Scheme (M.Sc.(IT) Programme) Effective from Year 2021-22

SEMESTER-I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
MS114	Software Engineering Paradigms	3	-	3	3	10	20	70	-	-	-	100
MS115	Framework And Application using .NET	3	3	6	6	10	20	70	15	15	70	200
MS116	Multi Paradigm Scripting using Python - I	3	3	6	6	10	20	70	15	15	70	200
MS117	Database Technologies	3	3	6	6	10	20	70	15	15	70	200
MS118-MS119	Elective-I	3	-	3	3	10	20	70	-	-	-	100
HS141.02 C HS105.02 C	Foreign Languages Academic Speaking and Presentation Skills	-	2	2	2	-	-	-	30		70	100
		15	11	26	26	500			400			900

**** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **CA730 Internet and Web Designing** course for others.**

Elective-I

1. MS118 Fundamentals of Computer Networks
2. MS119 Operating System Concepts

Teaching and Examination Scheme (M.Sc.(IT) Programme)
Effective from Year 2021-22

SEMESTER-II

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
MS217	Web Development using Open Source Technology	3	3	6	6	10	20	70	15	15	70	200
MS218	Software Testing	3	3	6	6	10	20	70	15	15	70	200
MS219	Object Oriented Programming using Java	3	3	6	6	10	20	70	15	15	70	200
MS220-MS222	Elective II	3	3	6	6	10	20	70	15	15	70	200
HS106.02 C	Academic Writing	-	2	2	2	-	-	-	30		70	100
	University Elective-I**	-	2	2	2	-	-	-	30		70	100
		12	16	28	28	400			600			1000

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **CA842 Mobile Application Development** course for others.

Elective-II

1. MS220 HTTP Web Service for Enterprise Application
2. MS221 Multi Paradigm Scripting using Python - II
3. MS222 Game Development - I

University Elective-I

No	Course Code	Course Name	Department/Faculty
1	EE782.01	Energy Audit and Management	Engineering
2	CE771.01	Project Management	Engineering
3	PT796.01	Fitness & Nutrition	Physiotherapy
4	MB651	Software based Statistical Analysis	Management
5	NR755	First Aid & Life Support	Nursing
6	OC733.01	Introduction to Polymer Science	Applied Science
7	MA771.01	Reliability and Risk Analysis	Mathematics
8	ME781.01	Occupational Health & Safety	Engineering
9	MA772.01	Design of Experiments	Mathematics
10	RD701.01	Introduction to Analytical Techniques	Applied Science
11	RD702.01	Introduction to Nanoscience And Technology	Applied Science
12	PH891	Community Pharmacy Ownership	Pharmacy
13	PH892	Intellectual Property Rights	Pharmacy
14	PSE55	Astrophysics, Space and Cosmos	Applied Science

Teaching and Examination Scheme (M.Sc.(IT) Programme)
Effective from Year 2021-22

SEMESTER-III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Tot al		Internal		Exter nal	Internal		Exter nal	
						Case Study	Tests		Term work	Tests		
MS320	Open Source Frameworks	3	3	6	6	10	20	70	15	15	70	200
MS321	Advanced Java	3	3	6	6	10	20	70	15	15	70	200
MS322	JavaScript Frameworks	-	3	3	3	-	-	-	15	15	70	100
MS323	Mobile Application technology	3	3	6	6	10	20	70	15	15	70	200
MS324-MS327	Elective-III	3	3	6	6	10	20	70	15	15	70	200
CL144.03C	Environmental Sciences	-	2	2	2	-		-	30		70	100
		12	17	29	29	400			600			1000

Elective-III

1. MS324 Game Development-II
2. MS325 Data Analytics
3. MS326 Ruby on rails
4. MS327 Internetworking with TCP/IP

Teaching and Examination Scheme (M.Sc. (IT) Programme)
Effective from Year 2021-22

SEMESTER-IV

Course Code	Course Title	Teaching Scheme				Internal	End Semester Examination		Total
		Contact Hours			Credit	Continuous Evaluation	Report	Presentation & Viva	
		Inst.	Industry	Total					
MS402	Dissertation/Project Work	2	28	30	30	200	200	400	800

TEACHING SCHEME & DETAILED SYLLABUS

FOR

**M.Sc. (IT) PROGRAMME
(1st SEMESTER)**

EFFECTIVE FROM ACADEMIC YEAR 2021-22

Teaching and Examination Scheme (M.Sc. (IT) Programme)
Effective from Year 2021-22

SEMESTER-I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
MS114	Software Engineering Paradigms	3	-	3	3	10	20	70	-	-	-	100
MS115	Framework And Application using .NET	3	3	6	6	10	20	70	15	15	70	200
MS116	Multi Paradigm Scripting using Python - I	3	3	6	6	10	20	70	15	15	70	200
MS117	Database Technologies	3	3	6	6	10	20	70	15	15	70	200
MS118-MS119	Elective-I	3	-	3	3	10	20	70	-	-	-	100
HS141.02 C HS105.02 C	Foreign Languages Academic Speaking and Presentation Skills	-	2	2	2	-	-	-	30		70	100
		15	11	26	26	500			400			900

**** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **CA730 Internet and Web Designing** course for others.**

Elective-I

1. MS118 Fundamentals of Computer Networks
2. MS119 Operating System Concepts

**MS114: Software Engineering Paradigms
(100 Marks)**

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures, the emphasis will be given on the basics of software engineering concepts and UML. Some of the agile software development concepts will be introduced during lectures. As part of modeling with UML, students will be given exposure to model real-life software projects.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours
		Theory
1	Introduction to software & software engineering	06
2	The software process	07
3	Introduction to Agile Development and Extreme Programming	07
4	Requirement Modeling: Scenarios, Information and Analysis classes	05
5	Requirement Modeling: Behavior, Patterns and WebApp	05
6	Software Maintenance and Risk Management	06

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I Introduction to software engineering Hours: 06

The nature of software, Software engineering layers, the software process generic framework activities, Umbrella activities of software engineering, Software Engineering Practices : Essence of practice, principles of software engineering.

Unit – II The Software Process Hours: 07

Prescriptive Process Models :: the waterfall model, Incremental software process models, evolutionary process models

Unit – III: Introduction to Agile Development and Extreme Programming Hours: 07

Introduction to Agility, Agility and cost of change, Agile process: Agility principles, Introduction to Extreme programming: XP values, XP process, Industrial XP practices, XP issues Overview of Agile process models : Adaptive software development, SCRUM, Dynamic system development method, Crystal, Feature drive development, Lean software development

Unit –IV: Requirement Modeling: Scenarios, Information and Analysis classes Hours: 05

Creating structural models: creating class diagrams, Developing object diagrams, examples

Unit –V: Requirement Modeling: Behavior, Patterns and WebApp Hours: 05

Creating behavioral models: creating use-case diagrams, Developing activity diagrams, Swim lane diagrams, introduction to sequence diagram, examples

Unit – VI: Software Maintenance and Risk Management Hours: 06

Software risks, risk identification, risk projection, risk mitigation, monitoring and management, software maintenance, business process reengineering, software reengineering, reverse engineering, restructuring, forward engineering.

Core Books:

1. Grady Booch, James Rumbaugh, Ivar Jacobson: The Unified Modeling Language User 2. Roger S. Pressman: Software engineering – A Practitioner's Approach, 7th Edition, ISBN: 978-007-126782-3, McGraw-Hill Publication.

Reference Books:

1. Ian Sommerville: Software Engineering, 8th Edition, ISBN: 978-81-317-2461-3, Pearson Education.
2. Guide, Addison Wesley. Jacobson, Booch, Rumbaugh, The Unified Software Development Process. Pearson Education, 1999.

Web References:

1. https://en.wikibooks.org/wiki/Introduction_to_Software_Engineering/Process/Methodology [Introduction to Software Engineering/Process/Methodology]
2. <https://www.uml-diagrams.org/> [UMLUnits]
3. <https://nptel.ac.in/courses/106101061/26> [Agile Software Development and Extreme Programming]
4. <https://nptel.ac.in/courses/Webcoursecontents/IIT%20Kharagpur/Soft%20Engg/pdf/m12L31.pdf> [Risk Management]

Course Outcomes: Upon successful completion of the course, the students will :

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C01 : Understand basics of Software Engineering

C02 : Develop understanding of various software process models

C03 : Understanding of the Agile software development concepts & extreme programming

C04 : Be able to model using structural diagrams & behavioral diagrams

C05 : Understanding the Risks associated with Software Development Process and managing them

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to software & software engineering	✓				
2	The software process		✓			
3	Introduction to Agile Development and Extreme Programming			✓		
4	Requirement Modeling : Scenarios, Information and Analysis classes				✓	
5	Requirement Modeling : Behavior, Patterns and WebApp				✓	
6	Software Maintenance and Risk Management					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	2	2	3	2	1	2	2	3	3
CO2	3	3	3	2	3	3	2	3	2	2	3	2	3	2
CO3	3	3	3	3	2	3	2	3	3	2	3	3	3	3
CO4	3	3	3	3	3	3	3	3	3	2	3	3	3	3
CO5	3	2	2	3	2	3	2	3	3	2	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS115: Framework and Applications using .NET
(200 Marks)

Contact Hours: 06

Pre-requisite: Familiarity with basic concepts of object oriented programming and Extensible Markup Language (XML) concepts.

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for advanced features of .Net framework and ASP.Net will be given. During Practical sessions, students will be required to develop Web Applications using concepts discussed during class.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Introduction to .NET framework and ASP.NET	04	36
2	Working with ASP.NET Applications	12	
3	Database Connectivity through ADO.NET	06	
4	Overview of SOA	04	
5	Reporting	05	
6	Advance ASP.NET	05	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit - I: Introduction to .NET framework and ASP.NET Hours:04

Introducing .NET Framework Enterprise vision of .NET , .NET Framework Component , .NET Framework Version Compatibility ,Core of .NET Framework: Application Services, Base Class Library and CLR Introducing Web Programming.

- Understanding Web server (IIS) and Web Client.
 - Basic of Http Request and Http Response.
 - Understand form Tag and Comparison between Get and Post
- Methods Introducing ASP and ASP.NET
 - Programming in ASP using visual studio
 - Programming in ASP.NET using visual
- studio Deploying ASP.NET Application
 - Compilation and execution of .NET Application
 - Dynamic Compilation

Unit - II: Working with ASP.NET Applications Hours:12

- ASP.NET Page Life Cycle
- Structure of an ASP.NET Page: ASPX Page, Code behind File, WebConfig and machine config
- Develop Web Form
 - Create User Interface using Standard Controls, Rich Web Control, Navigation Controls and Login Control.
 - Working with properties and events of controls
 - Validate User Input using Validation Controls.
 - Concept of custom control.
- Concept of MasterPage and Nested MasterPage
- State Management Techniques
 - Client side: ViewState, Hidden Field, Cookie
 - Server Side: ApplicationState, Session, Output cache
 - Application Tracing, Error Handling and Responding to Errors
 - Try Catch Final
 - Custom Error Page

Unit - III: Database connectivity through ADO.NET Hours:06

- Introduction and Evolution of ADO.NET.
- Binding data to web controls and data controls.
- ADO.NET Architecture:

- Connection Oriented: Understanding of Connection, Command, DataReader object.
- Connection Less: Understanding of DataAdapter, DataSet, DataTable and DataView object.
- Working with XML – Overview of XML Classes and using XML with datasets.

Unit - IV: Overview of SOA Hours:04

- Overview of Service Oriented Architecture
- Service Provider ○ Service Consumer ○ Service ○ Service Description
 - SOAP
 - UDDI
- Building Web Service in ASP.NET ○ Deploying, Publishing and Consuming Web service.

Unit - V: Reporting Hours:05

Introduction to Crystal Report, Crystal Reports Architecture, ReportViewer Control, Object Model, Understanding Reporting Control

Unit - VI: Advance ASP.NET Hours:05

- Introduction to ASP.NET AJAX.
- ASP.NET AJAX Control Toolkit Extender and controls.

Adjusting the Web Content, Validating Controls, Working with Menu, list and pop-ups.

Introduction to MVC Architecture in ASP.NET, Developing web application in MVC Architecture.

Core Books:

1. Stephon Walther: ASP.Net Unleashed, BPB publication.
2. Kogent Solutions Inc.: ASP.Net 4.5 Covers C# 2012 & VB 2012 Black book, Dreamtech press.
3. Mridila Parihar, Essam Ahmed: ASP .Net Bible, Wiley, 2004.

Reference Books:

1. Bill Evjen, Scott Hanselman, Devin Rader: Professional ASP.Net 3.5 (sp1) In C# and VB, Wiley India Edition.
2. Mesbah Ahmed, Chris Garrett, Jeremy Faircloth, Chris Payne: ASP.Net Programming. Developer's Guide, Dreamtech, 1 st Edition 2002.
3. A. Russell Jones, Mike Gunderloy: .Net Programming 10-Minute Solutions, BPB Publications.
4. Greg Buczek: ASP.Net Developer's Guide, Tata McGraw Hill 4th Edition, 2005.
5. Greg Buczek: ASP.Net Tips & Techniques, Tata McGraw Hill Edition, 2002.
6. Bolton, Justin Langford, Glenn Berry, Gavin Payne, Amit Banerjee, Rob Farley: Professional SQL Server 2012 internals and trouble shooting, Wiley India publication, October,2012.

Web References:

1. <http://msdn.microsoft.com/en-us/aa336522.aspx> [For Unit 1 and to download software]
2. <http://www.asp.net/> [For Unit1 & 2 and to download software]
3. <http://www.aspfree.com/> [Forum for discussion on ASP.NET]
4. <http://www.devx.com/dotnet> [To read latest published articles/news]

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5. myweb.sabanciuniv.edu/gulsend/files/2010/03/intro.ppt [Lecture Note for Unit1 and 2]
6. www.cs.odu.edu/~mukka/cs795sum08/Lecturenotes/Day3/ado.ppt [Lecture Note for Unit 3]
7. <http://www.codeproject.com/Articles/142064/Step-by-Step-Creation-of-Crystal-Reportusingits> [Lecture Note for Unit 5]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Understanding of .Net Framework

C02 : Students will be able to develop, test and deploy dynamic web applications

C03 : Understanding of database connectivity and creation of application

C04 : Students will be familiar with the concepts of SOA

C05 : Students will be familiar with the reporting tools

C06 : Students will be familiar with the concepts of AJAX and other advance .net technology

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1	Introduction to .NET framework and ASP.NET	✓					
2	Working with ASP.NET Applications		✓				
3	Database Connectivity through ADO.NET			✓			
4	Overview of SOA				✓		
5	Reporting					✓	✓
6	Advance ASP.NET						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	2	3	2	3	-	3	2	-	-	-	-	-	2
CO2	3	3	3	3	3	3	3	3	2	3	3	3	3	3
CO3	3	2	2	2	2	2	3	2	-	2	-	2	2	-
CO4	3	2	2	2	3	3	2	3	-	2	3	2	3	2
CO5	3	3	3	2	3	-	3	-	3	3	2	2	-	3
CO6	3	3	3	3	3	3	3	3	3	3	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS116 : Multi Paradigm Scripting using Python - I
(Marks 200)

Contact Hours: 06

Pre-requisite: Programming principals and Logic Development, Fundamentals Concepts of Programming Language, Object Oriented Programming Using C++

Methodology & Pedagogy: Theory sessions are required to address computational power of python through its ability to deploy programs using functional, object oriented and web based aspect. Practical sessions demonstrate the implementation of the concepts which are taught during the theory sessions. Case study will help the students to come out with one working module in any of the paradigm through python.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Introduction to Python in context of Multi Paradigm Script	08	36
2	Getting Started with Python	08	
3	Python Objects, Mapping, Set & Iterative Programming	05	
4	Python functional Programming	05	
5	Object Oriented Programming using Python	05	
6	Advanced Programming using Python	05	

Total Hours (Theory): 36

Total Hours(Lab): 36

Total Hours: 72

Detailed Syllabus:

Unit – I: Introduction to Python in context of Multi Paradigm Script Hours: 08

Overview of programming Paradigms-Imperative, Functional, logic and object oriented Introduction to Python Programming Language, Downloading and Installing, Running Python, Python Documentation. Features of Python and its origin.

Unit – II: Getting Started with Python Hours: 08

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Input through standard input device, Generating Output on console, Operators in python, how to use Numbers in python, String and its variations in python, Comment statement in python , Various Statement & its Syntax, Python Identifiers, Basic Programming Style of Python, Memory Management scheme adopted in Python, Conditional & Branching Statement in python.

Unit – III: Python Objects, Mapping, Set & Iterative Programming: Hours: 05

Python objects and its variations: List, Tuple & Set

Different looping constructs available in Python: While, for, continue, break and pass statement Special Python object like Dictionary, nesting of one object into another/same object, and other built in functions for list, tuple, set and dictionary.

Unit – IV: Python functional Programming: Hours: 05

What are functions, calling functions, creating functions, passing functions, formal arguments, variable length arguments, default arguments, returning values from the functions, returning multiple values from the functions, functional programming, variable scope and recursion.

Unit – V: Object Oriented Programming using Python: Hours: 05

Basics of class, object and instance. Class level attribute and instance level attribute. Constructor and other magic methods. Bound and unbound methods. Built in functions for python class and objects.

Unit – VI: Advanced Programming using Python: Hours: 05

Class mappings, inheritance, Introduction to client server programming. Introduction to Python GUI programming using Tkinter

Core Books:

1. Wesley J. Chun : Core Python Programming, 2nd edition, Pentice Hall,2006.
2. Megnus Lie Hetland : Beginning Python from novice to professional, 2nd edition, Apress,2009.

Reference Books:

1. Mark Lutz : Programming Python, 4th Edition,O'reilly, 2011 .
2. Dusty Philips: Python 3 Object oriented Programming , PACKT publishing, 2010.
3. Steve Holden: Python Web Programming, 1st edition,2002.

Web References:

1. http://people.cs.aau.dk/~normark/prog3-03/html/notes/paradigms_themes-paradigmo_verview-section.html [Overview of Programming Paradigms]
2. www.tutorialspoint.com/python [For tutorials of Python]
3. <https://docs.python.org> [Python Documentation]
4. <https://www.python.org/about/gettingstarted/> [For beginners of Python]
5. http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-189-a-gentl_eintroduction-to-programming-using-python-january-iap-2008/ [Python Materials]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Install and run the Python interpreter, Create and execute Python Program

C02 : Adequately use standard programming constructs: repetition, selection, functions, composition, modules, aggregated data (arrays, lists, etc.)

C03 : Explore Python's object-oriented features

C04 : Able to use library software for building a graphical user interface or web application

C05 : Write Python functions to facilitate code reuse

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Python in context of Multi Paradigm Script	✓				
2	Getting Started with Python	✓	✓			
3	Python Objects, Mapping, Set & Iterative Programming		✓			
4	Python functional Programming		✓			✓
5	Object Oriented Programming using Python			✓		
6	Advanced Programming using Python				✓	

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	2	2	2	1	-	-	-	-	-	-	-	3	3
CO2	3	2	2	3	2	2	-	3	-	-	-	-	2	3
CO3	-	-	3	3	2	-	-	-	-	-	-	-	2	3
CO4	3	2	2	-	2	2	-	3	-	-	-	-	3	3
CO5	2	-	-	-	2	-	3	3	-	-	-	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS117 : Database Technologies
(200 Marks)

Contact Hours: 06

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures, illustrations of certain complex real world applications, which emphasis the use of advanced concepts of databases, will be discussed. The fundamental concepts regarding database development activities, various database management systems and other advanced issues in database management will also be discussed. In addition, there may be announced or unannounced quizzes/assignments. During Practical sessions, students will be required to carry out case studies using the concepts and techniques they have learnt during theory sessions.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Database concepts	06	36
2	Relational model	08	
3	Transaction and concurrency control	06	
4	Query optimization	06	
5	Distributed and object relational model	05	
6	Data Warehousing, Data Mining	05	

Total Hours (Theory): 36

Total Hours(Lab): 36

Total Hours: 72

Detailed Syllabus:

Unit – I: Database concepts: 06

Database- Applications of database, Purpose of Database Systems, View of Data, Database Languages, Types of Database Models, Relational Databases, Database Design, Database Architecture, Database Users and Administrators

Unit – II: Relational Models Hours: 08

Structure of Relational Databases, Database Schema, Keys, Schema Diagrams, Relational Query Languages, Relational Operations. Overview of the SQL Query Language, SQL Data Definition, Basic Structure of SQL Queries, Basic Operations, Set Operations, Null Values, Aggregate Functions, Nested Subqueries, Modification of the Database, Join Expressions, Views, Transactions, Integrity Constraints, SQL Data Types and Schemas

Unit – III: Transaction and concurrency control Hours: 06

Transaction properties, Transaction log, Transaction States, Concurrency control, Locking methods for concurrency control, Deadlock handling, detection and resolution, Types of Database recovery (Forward recovery, Backward recovery, Media recovery), Recovery techniques (Deferred Update, Immediate update, Shadow Paging).

Unit – IV: Query optimization Hours: 06

Heuristics for query optimization, join strategies, Database Workloads, Tuning Decisions, DBMS Benchmarks, Query Evaluation Plans, System Catalogue in RDBMS

Unit – V: Distributed and object relational model Hours: 05

Distributed database concepts, Data fragmentation, Replication and allocation techniques for distributed database design, Introduction to ORDBMS, Database design for an ORDBMS, An overview of SQL3, a comparison of RDBMS, OODBMS and ORDBMS.

Unit – VI: Data Warehousing, Data Mining Hours: 05

Introduction to data warehousing, components, characteristics, advantages and limitations of data warehouses, Introduction to data mining, goals of data mining, data mining tools and applications

Core Books:

1. Raghu Ramakrishnan, Johannes Gehrke: Database Management Systems, McGraw Hill Publication.
2. Ramez Elmasri, Shamkant B. Navathe: Fundamentals of Database Systems, 5th Edition, Pearson Publication.
3. S.K. Singh: Database Systems, Concepts, Design and Applications, 1st Edition, Pearson Publication.

Reference Books:

1. Abraham Silberschatz, Henry F. Koeth, S. Sudarshan: Database System Concepts, 6th edition, McGraw Hill Publication.

Web References:

1. <http://www2.amk.fi/digma.fi/www.amk.fi/opintojaksot/0303011/1146161367915/1146161783414/1146163033699/1146163141034.html> [Database system concepts]
2. <https://cs.nyu.edu/courses/fall09/G22.2434-001/index.html> [Advanced Database System]
3. [Http://infolab.usc.edu/csci585/Spring2010/den_ar/ordb.pdf](http://infolab.usc.edu/csci585/Spring2010/den_ar/ordb.pdf) [Object Relational Database]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : To understand basic of data, database systems and develop normalization concepts using relational model.

C02 : Getting deep knowledge of underlying transaction concepts.

C03 : Learning query processing and query optimization.

C04 : To gain understanding of various databases

C05 : To learn data warehousing and data mining concepts.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Database concepts	✓				
2	Relational model	✓				
3	Transaction and concurrency control		✓			
4	Query optimization			✓		
5	Distributed and object relational model				✓	
6	Data Warehousing, Data Mining					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	-	-	-	1	-	-	-	-	3	3
CO2	2	2	1	1	-	1	-	-	-	-	-	-	2	3
CO3	1	3	3	2	-	1	3	-	-	-	-	-	2	3
CO4	2	1	2	2	2	1	2	2	-	-	-	-	3	3
CO5	1	1	2	3	1	-	2	-	-	-	-	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**MS118: Fundamentals of Computer Networks
(100 Marks)**

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures the emphasis will be given on the Introduction to the computer network basics and terminologies used in computer networking. The teacher will also discuss different reference models and the working of each layer of models. The teacher will discuss different design issues related to layers of model and networking devices and networking protocols.

Outline of the Course:

Unit No.	Title of Unit	Minimum Number of Hours Theory
1	Basics of Computer Network	05
2	Network Communication Concepts	07
3	Reference Models	05
4	Issues in Data Link Layer Design	08
5	Issues in Network Layer Design	06
6	Network Protocols & Devices	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit - I: Basics of Computer Network Hours: 05

Definition of Computer Network, Goals of Computer Network: For Businesses, For People, Advantages & Disadvantages of Computer Network, Applications of Computer Network, Components of Computer Network, Types of Networks, Network Topologies.

Unit - II: Network Communication Concepts Hours: 07

Parallel & Serial Communication, Synchronous & Asynchronous Communication. Modes of Communication: Simple, Half-Duplex, Full-Duplex, Multiplexing, De-Multiplexing: FDA, TDM.

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Communication Media: Guided Media, Unguided Media. Switching Techniques: Circuit Switching, Message Switching, Packet Switching.

Unit - III: Reference Models Hours: 05

OSI Reference Model: Architecture, Function of Each Layer, TCP/IP Reference Model: Architecture, Function of Layers. Comparison of OSI & TCP/IP Models.

Unit - IV: Issues in Data Link Layer Design Hours: 08

Services provided to the Network Layer, Framing, Error Control (error detection and correction code), Flow Control, Data Link Layer in the Internet (SLIP, PPP). MAC sub layer: CSMA/CD/CA, IEEE standards (IEEE 802, IEEE802.3, IEEE 802.4, IEEE 802.5, IEEE 802.11).

Unit - V: Issues in Network Layer Design Hours: 06

Routing Algorithms: Optimality Principle, Shortest Path Routing, Congestion Control Policies, Concept of Internetworking.

Unit - VI: Network Protocols & Devices Hours: 05

Protocol Hierarchy, Network Protocols: SMTP, FTP, MIME, POP, Telnet, DNS, DHCP, ARP, RARP. Network Devices: Hub, Switch, Repeater, Router, Gateway, Bridge.

Core Books:

1. Behrouz A. Forouzan: Data Communications and Networking, Tata McGraw-Hill, 4th Edition, 2006.

Reference Books:

1. Andrew S. Tanenbaum: Computer Networks, Prentice Hall, 4th Edition Person, 2013.
2. William Stallings, Data and Computer Communication, 8th Edition, Pearson Education, 2007.

Web References:

1. <http://www.cs.iit.edu/~cs549/lectures/CNS-1.pdf> [For different books and chapters of the networking technology].
2. <http://nptel.ac.in/> [Online Lectures].
3. http://www.tutorialspoint.com/data_communication_computer_network/index.html [Course Material, Notes].

Course Outcomes: Upon successful completion of the course, the students will :

- C01 : Better understanding of basic networking concepts and terminologies
- C02 : Be familiar with communication concepts
- C03 : To understand the functions of reference models.
- C04 : Understand and solve issues of data link layer.
- C05 : Understand and solve issues of network layer.
- C06 : Understand various networking devices and protocols.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1	Basics of Computer Network	✓					
2	Network Communication Concepts		✓				
3	Reference Models			✓			
4	Issues in Data Link Layer Design				✓		
5	Issues in Network Layer Design					✓	
6	Network Protocols & Devices						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3	3	-	1	1	-	-	1	1	-	3	-	2
CO2	3	2	3	2	1	1	-	-	-	1	-	3	1	3
CO3	3	3	2	2	1	2	2	1	-	2	1	2	1	3
CO4	3	2	3	2	1	1	2	1	-	1	2	1	1	3
CO5	2	3	3	2	1	2	3	1	-	2	1	2	1	2
CO6	2	3	3	2	1	2	3	1	-	2	1	2	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**MS119: Operating System Concepts
(100 Marks)**

Contact Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: The course will start with a brief introduction of operating system, and then cover the major components of most operating systems. This discussion will cover the tradeoffs that can be made between performance and functionality during the design and implementation of an operating system. Particular emphasis will be given to these major OS subsystems: process management (processes, threads, CPU scheduling, synchronization, and deadlock), memory management (segmentation, paging, swapping), file systems and System protection.

Outline of the Course:

Unit No.	Title of Unit	Minimum No of Theory Hours
1	Introduction	05
2	Process Management and Scheduling	07
3	Concurrency Control and Deadlock	06
4	Memory Management	07
5	File System and Storage Management	06
6	I/O Management and System Protection	05

Total Hours (Theory): 36

Total: 36

Unit – I: Introduction Hours: 05

Overview of Computer System and Architecture, Introduction to Operating Systems – Structure & Architecture, Functions of Operating Systems, Types of Operating systems, Operating system Interface, OS services, System calls and its types, Operating Systems Structure, design and its implementation, Virtual Machines.

Unit – II: Process Management and Scheduling Hours: 07

Process-concept, Process Control Block, Process Trace, Dispatcher, Process State Models- Two State, Five State, Seven State, Process Vs Thread, Processor scheduling and its algorithms

Unit – III: Concurrency Control and Deadlock Hours: 06

Semaphore, Producer Consumer Problem Using Semaphore, Binary Semaphor, Monitor, Producer Consumer Problem using Monitor
Deadlock, Conditions of Deadlock, Deadlock Prevention, Deadlock Avoidance

Unit – IV: Memory Management Hours: 07

Memory management requirements, Swapping, contiguous memory allocation, Fragmentation, Paging, Structure of Page table, Segmentation
Virtual memory management: Demand paging, Page replacement algorithms, Allocation of frames, Thrashing, Copy-On-Write

Unit – V: File System and Secondary Storage Hours: 06

File concepts, access methods, Directory structure, Mounting, File sharing, Protection, File system structure, File system implementation, Directory implementation, allocation methods, free space management, Efficiency and performance, recovery.
Overview of Storage devices, Disk structure, Disk attachment, Disk scheduling, Disk Management, wapSpace Management, RAID, Tertiary Storage Structure

Unit – VI: I/O Management and System Protection Hours: 05

I/O management: I/O hardware, Application I/O Interface, Kernel I/O subsystems, Streams, Device drivers
System protection: Protection goals, Access matrix, access control, Security problem, threats, Introduction to security measures.

Core Books:

1. William Stallings: Operating Systems Internals and Design Principles, 6th Edition, PHI 2009.
2. Silberschatz, A., Peter B. Galvin and Greg Gagne: Operating System Principles, 8th Edition Wiley – Indian Edition, 2009.
3. Andrew S. Tanenbaum: Modern Operating Systems, Prentice Hall of India Pvt. Ltd, 2003.

Reference Books:

1. Kenneth Rosen, Douglas Host, James Farber and Richard Rosinski: The Complete Reference, Tata McGraw Hill, 1999.
2. Mark G. Sobell: A Practical Guide to Linux, Pearson Education, 1997.
3. Harvey M. Deitel: Operating Systems, 2nd Edition, Pearson Education Pvt. Ltd, 2002.

Web References:

1. <http://faculty.simpson.edu/lydia.sinapova/www/cmsc335/StallingsPPT.htm> [PPTs of Operating System Concepts]

2. http://www.tutorialspoint.com/operating_system/operating_system_tutorial.pdf [OS Tutorials]
3. <http://stst.elia.pub.ro/news/SO/Modern%20Operating%20System%20-%20Tanenbaum.pdf> [Reference book of OS]

Course Outcomes: Upon successful completion of the course, the students will:

C01 : Be familiar with Operating Systems, its types and services

C02 : Be familiar with Process and Threads and Deadlock

C03 : Be familiar with Memory Allocation Techniques, Virtual Memory and Paging

C04 : Get acquainted with file structures and storage devices and system protection

C05 : Be able to understand the principles and internals of Operating System.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction	✓				✓
2	Process Management and Scheduling		✓			✓
3	Concurrency Control and Deadlock		✓			✓
4	Memory Management			✓		✓
5	File System and Secondary Storage				✓	✓
6	I/O Management and System Protection				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	-	-	-	1	-	-	-	-	3	3
CO2	2	2	1	1	-	1	-	-	-	-	-	-	2	3
CO3	1	3	3	2	-	1	3	-	-	-	-	-	2	3
CO4	2	1	2	2	2	1	2	2	-	-	-	-	3	3
CO5	1	1	2	3	1	-	2	-	-	-	-	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS141.02 A: FOREIGN LANGUAGES (French)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02/01	--	30/15	02
Marks	--	100	--	100	

Pre-requisite courses:

- French Language Studies- Introduction (Coursera)

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to French Language	08
2.	Grammar: Articles, Tense, Forms, Numbers, Verbs, Days, Months, Family	08
3.	Grammar : Adjectives, Adverbs, Interrogative Forms, Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs	08
4.	Grammar: Prepositions, Conjunctions, Tenses, Colours, Vegetables, Fruits, Shapes, Verbs	06
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

1.	Introduction to French Language	08 Hours	28%
	Facts and figures about French Language; Basic French Linguistics- * Alphabets * Accents * Liaison * Nasalization French Culture, Differ between French and English; Grammar- Subject Pronoun, Verbs: (être, avoir, habiter, regarder, manger ... “er” verb), Form of address, Numbers (1 to 20), Nouns and plurals of nouns, The expression: C’est, Il y a; Presentation: -1) Self-Introduction-2) Question and answering; Dialogue		
2.	Grammar: Articles, Tense, Forms, Numbers, Verbs, Days, Months, Family	08 Hours	28%
	Grammar- Definite articles, Indefinite articles, Present tense (Positive Forms, Negative Forms), Numbers (21 to 100, 100-1000), Days, Months, Family, Verbs: (aller, venir, finir, pouvoir, vouloir ...		

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	"ir" verb); Social Links -1), My family & relations 2) Appointments 3) Gathering information from someone; Dialogue		
3.	Grammar : Adjectives, Adverbs, Interrogative Forms, Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs	08 Hours	28%
	Grammar - Common Adjectives, Comparative Adjectives, Common Adverbs, Interrogative Forms, The expression: "On", Directions, Countries, Nationalities, Seasons, Weather, Professions, Verbs: (Prendre, Apprendre, Comprendre, faire ... "re " verb); Work , Study and Travel -1) Job/ Profession 2) Ticket Reservation (At Bus/At Railway/At Airport); Dialogue		
4.	Grammar: Prepositions, Conjunctions, Tenses, Colours, Vegetables, Fruits, Shapes, Verbs	06 Hours	26%
	Grammar -1) Common Prepositions 2) Common Conjunctions 3)Past Tense 4) Future Tense 5) Colors ,Shapes, Animals ,Vegetables, Fruits 6) Verbs: ("er", "ir","re" etc...); Food & Shopping -1) Buy a vegetables and fruits 2) Any Conversation between Customer and Vendor (At Mall/At Restaurant / At Market); Dialogue		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Gain basic communication skills in French language with preliminary understanding of grammar
CO2	Develop vocabulary required to speak about him/herself and his/her immediate environment.
CO3	Become capable of interacting in simple ways, to ask simple questions to get necessary information, to reply simple questions.
CO4	Become capable of understanding and using simple instructions in their personal, academic and professional environments.
CO5	Develop skills and intelligences to function in multi-disciplinary and cross-cultural work environment.
CO6	Practice new global trends in communication in multiple perspectives at personal, professional, and social level.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO6	-	-	-	-	-	-	3	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

- **Text book:**

1. Complete French: All-In-One, McGraw-Hill, Amazon
2. Best for Grammar: Easy French Step-by-Step, McGraw-Hill, Amazon

- **Reference book:**

1. Basic French: McGraw-Hill, Amazon
2. French Grammar for Beginners, Amazon

- **Web material:**

1. <https://alison.com/course/french-language-studies-introduction>
2. <https://alison.com/course/basic-french-language-skills-for-everyday-life-revised-2017>
3. <http://www.bbc.co.uk/languages/french/>
4. <https://www.loecsen.com/en/learn-french>
5. <https://www.youtube.com/watch?v=ujDtm0hZyII>

HS105.02 C: ACADEMIC SPEAKING AND PRESENTATION SKILLS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	02
Marks	--	100	--	100	

Pre-requisite courses:

- Beginner/Intermediate level language proficiency

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Foundations of Advance Communication	04
2.	Art of Conversation	06
3.	Science of Power Speaking	06
4.	Academic Speaking Application – Part I	08
5.	Academic Speaking Application – Part II	06
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

1.	Foundations of Advance Communication	04 Hours	14%
	Meaning and Definition of Advance Communication; Advance Communication in Digital, Social, Mobile World; Strategies for Advance Communication; Meaning and Concept of Academic Language; High Frequency Academic Vocabulary		
2.	Art of Conversation	06 Hours	20%
	Describing people, places and things; Expressing opinions; Making suggestions; Persuading someone; Interpreting and Summarizing		

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3.	Science of Power Speaking	06 Hours	20%
	Phonemes, Word Stress, Pronunciation, Intonation, Pause, Register, Fluency, Prosody, Lexical Range		
4.	Academic Speaking Application – Part I	08 Hours	26%
	Art of Oratory, Formal Presentation, Speech Analysis – Decoding Best Speeches		
5.	Academic Speaking Application – Part II	06 Hours	20%
	Job Interview, Group Discussion, Meeting		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	understand and demonstrate advance communication and academic speaking skills
CO2	demonstrate ability to communicate in diverse situations
CO3	activate and extend their linguistic and communicative competence
CO4	demonstrate the formal presentation skills
CO5	demonstrate performing ability at group discussion and personal interview

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	2	-	3	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	3	-	3	-	-	-

Recommended Study Material:

- **Text book:** -

- **Reference book:**

3. *Business Communication Today* (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina

4. *Effective Speaking Skills* by Terry O' Brien
5. *Speak Better Write Better* by Norman Lewis
6. *Well Spoken: Teaching Speaking to All Students* by Erik Palmer
7. *Let Us Hear Them Speak : Developing Speaking – Listening Skills in English* by Jayshree Mohanraj (Publisher – Sage Publication)
8. *The craft of scientific presentations: Critical steps to succeed and critical errors to avoid.* New York: Springer by Michael Alley
9. *Presentation Skills in English* by Bob Dignen (Publisher: Orient Black Swan)

- **Web material:**

1. TED Talk : How to speak so that people want to listen
https://www.ted.com/talks/julian_treasure_how_to_speak_so_that_people_want_to_listen?language=en
2. TED Talk: The 110 techniques of communication and public speaking
https://www.ted.com/talks/david_jp_phillips_the_110_techniques_of_communication_and_public_speaking

TEACHING SCHEME & DETAILED SYLLABUS

FOR

**M.Sc. (IT) PROGRAMME
(2nd SEMESTER)**

EFFECTIVE FROM ACADEMIC YEAR 2021-22

Smt. Chandaben Mohanbhai Patel Institute of Computer Applications

Teaching and Examination Scheme (M.Sc.(IT) Programme) Effective from Year 2021-22

SEMESTER-II

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
MS217	Web Development using Open Source Technology	3	3	6	6	10	20	70	15	15	70	200
MS218	Software Testing	3	3	6	6	10	20	70	15	15	70	200
MS219	Object Oriented Programming using Java	3	3	6	6	10	20	70	15	15	70	200
MS220-MS222	Elective II	3	3	6	6	10	20	70	15	15	70	200
HS106.02 C	Academic Writing	-	2	2	2	-	-	-	30		70	100
	University Elective-I**	-	2	2	2	-	-	-	30		70	100
		12	16	28	28	400			600			1000

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **CA842 Mobile Application Development** course for others.

Elective-II

1. MS220 HTTP Web Service for Enterprise Application
2. MS221 Multi Paradigm Scripting using Python - II
3. MS222 Game Development - I

University Elective-I

No	Course Code	Course Name	Department/Faculty
1	EE782.01	Energy Audit and Management	Engineering
2	CE771.01	Project Management	Engineering
3	PT796.01	Fitness & Nutrition	Physiotherapy
4	MB651	Software based Statistical Analysis	Management
5	NR755	First Aid & Life Support	Nursing
6	OC733.01	Introduction to Polymer Science	Applied Science
7	MA771.01	Reliability and Risk Analysis	Mathematics
8	ME781.01	Occupational Health & Safety	Engineering
9	MA772.01	Design of Experiments	Mathematics
10	RD701.01	Introduction to Analytical Techniques	Applied Science
11	RD702.01	Introduction to Nanoscience And Technology	Applied Science
12	PH891	Community Pharmacy Ownership	Pharmacy
13	PH892	Intellectual Property Rights	Pharmacy
14	PSE55	Astrophysics, Space and Cosmos	Applied Science

**MS217 : Web Application Development using Open Source Technology
(200 Marks)**

Contact Hours: 06

Pre-requisite: Basic Knowledge of HTML and Database Technologies.

Methodology & Pedagogy: During theory and practical sessions, students able to install & configure PHP and prerequisite software(s). Also, student will be emphasized to develop dynamic web applications.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Introduction to Open Source Software and their Configuration	04	36
2	Basics of PHP	07	
3	Working with Arrays and Objects in PHP	07	
4	HTML Forms with PHP	06	
5	Accessing Relational Databases using PHP	07	
6	Responsive Web Application Development using AJAX and PHP	05	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit – I: Introduction to open source software and their Configuration. Hours: 04

Overview of Open Source Software, Widely used Open Source Products, Development Philosophy, Open source vs. Closed Source, Open Source Technology Importance, Installation & Configuration of PHP, Introduction to PHP.

Unit – II: Basics of PHP. Hours: 07

PHP language Basics: Lexical Structure, Data types, Variables, Expressions and Operators, Control and Looping statements. Functions: Function Definition, Function Parameters, Returning Values. Strings: Usages and String Functions.

Unit – III: Working with Arrays and Objects in PHP. Hours: 07

Arrays: Types of Arrays and its Usages, Array functions.

Objects: Declaring Class, Properties, Methods, Exception Handling, Examples.

Unit – III: HTML Forms with PHP. Hours: 06

Working of HTML with PHP, Capturing data with PHP Using HTML Form Elements, Super Global Variables, Processing User Input, Handling and Avoiding Errors, Working with Regular Expressions, State Management Techniques in PHP.

Unit – IV: Accessing Relational Databases using PHP. Hours: 07

Using PHP to access Databases, Relational Databases and SQL, PHP Data Objects (PDO), MySQL Object Interface.

Unit VI: Responsive Web Application using AJAX and PHP Hours: 05

Introduction to AJAX, PHP and AJAX Example, AJAX Suggest and Autocomplete, AJAX Data Grid.

Core Books:

1. Kevin Tatroe, Peter MacIntyre, Rasmus Lerdorf: Programming PHP - Creating Dynamic Web Pages, 3rd Edition, Kindle Edition, O'REILLY Publication.
2. Christian Darie, Brinzarea Bogdan, Filip Chereches-Tosa, Mihai Bucicia: AJAX and PHP: Building Responsive Web Applications, Kindle Edition, Packt Publishing.
3. Matt Doyle: Beginning PHP 5.3, Wrox Publication, 2010.

Reference Books:

1. David Sklar and Adam Trachtenberg: PHP CookBook - Solutions and Examples for PHP Programmers, 3rd Edition, O'REILLY Publication.

Web References:

1. <https://www.w3schools.com/php/default.asp> [For Basics of PHP]
2. <https://opensource.com/resources/what-open-source> [For Open Source Software] 3. <https://www.tutorialspoint.com/php/index.htm> [For Practical Examples]

Course Outcomes: Upon successful completion of the course, the students will:

- CO1 : Understand the importance of open source technology.
- CO2 : Understand the working with PHP and its elements.
- CO3 : Be able to handle HTML Form and process user input using PHP.
- CO4 : Be able to develop interactive and dynamic web based application using PHP and MySQL
- CO5: Be able to develop the responsive web applications

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to open source software and their Configuration PHP	✓				
2	Basics of PHP		✓		✓	
3	Working with Arrays and Objects in PHP		✓		✓	
4	HTML forms with PHP			✓	✓	
5	Accessing Relational Databases using PHP			✓	✓	
6	Responsive Web Application using AJAX and PHP				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	1	1	1	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	2	2	3	3	2	2	2	2	2	1	1	1	3	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3	3	2
CO5	3	3	3	3	3	3	3	3	3	2	3	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS218 : Software Testing
(200 Marks)

Contact Hours: 06

Pre-requisite: Software Engineering.

Methodology & Pedagogy: The theory sessions will cover the software testing concepts and practices that support the production of quality software. The practical sessions will cover the application of testing techniques at various levels of testing using manual and automated testing tools. This is **Outline of the Course:**

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Basic concepts of Software Testing	05	36
2	White Box Testing Techniques	05	
3	Black Box Testing Techniques	07	
4	Levels of Testing	07	
5	Test Planning and Design	06	
6	Test Execution and Measurements	06	

Total Hours (Theory): 36

Total Hours(Lab): 36

Total Hours: 72

Detailed Syllabus:

Unit – I: Basic concepts of Software Testing Hours: 05

Introduction to testing and its importance in software development, software testing role, responsibilities and principles, concept of quality, Verification and Validation, Design test case, Introduction to Static Techniques and Dynamic test case design techniques.

Unit – II: White Box Testing Techniques Hours: 05

Overview of White Box, Control flow testing: Statement Coverage Testing, Branch Coverage Testing, Path Coverage Testing, Conditional Coverage Testing
Data flow testing: Data Flow Anomaly, Overview of Dynamic Data Flow Testing, Data Flow Graph, Data Flow Terms, Data Flow Testing Criteria.

Unit – III: Black Box Testing Techniques Hours: 07

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Overview of Black Box: Equivalence Class Partition, Boundary Value Analysis, Pairwise Technique, Cause Effective Graph, Decision Table.

Unit – IV: Levels of Testing

Hours: 07

Introduction to functional and non functional testing

Functional Testing : Unit Testing, Integration Testing, System Testing, User Acceptance Testing.

Sanity/Smoke Testing, Regression Test.

Non Functional Testing: Performance Testing, Scalability Testing, Interoperability Testing, Security Testing, Recovery Testing, Basic Testing.

Unit – V: Test Planning and Design

Hours: 06

Requirements Analysis/Design : Understand the requirements, Prepare Traceability Matrix Test Planning: Object, Scope of Testing, Schedule, Approach, Roles & Responsibilities, Assumptions, Risks & Mitigations, Entry & Exit Criteria, Test Automation, Deliverables. Test Cases Design: Write Test cases, Review Test cases, Test Cases Template, Types of Test Cases, Difference between Test Scenarios and Test Cases.

Test Environment setup: Understand the SRS, Hardware and software requirements, Test Data

Unit – VI: Test Execution and Measurements

Hours: 06

Test Execution: Execute test cases, Defect Tracking and Reporting: Types of Bugs, Identifying the Bugs, Bug/Defect Life Cycle, Reporting the Bugs, Severity and priority.

Test Closure: Criteria for test closure, Test summary report

Test Metrics: Test Measurements, significance of Test Metrics, Metric Life Cycle, Types of Manual Test Metrics.

Core Books:

1. Sagar Naik, Piyu Tripathy: Software Testing and Quality Assurance, Theory and Practice, Wiley, 2008.
2. Paul C. Jorgensen : Software Testing: A Craftsman's Approach, 4th Edition by , CRC press, Taylor and Francis Group, 2014
3. Roger S Pressman: Software Engineering – A Practitioner's Approach, 7th Edition, McGRAW HILL International Edition, 2010.

Reference Books:

1. Mauro Pezze, Michael Young: Software testing and Analysis- Process, Principles and Techniques, Wiley India, 2012.
2. Boris Beizer: Software Testing Techniques: 2nd Edition, Van Nostrand Reinhold, 1990.
3. Daniel Galin: Software Quality Assurance, Pearson Education, 2004.
4. Ron Patton: Software Testing, Pearson Education, 2005.

Web References:

1. <http://tryqa.com/what-is-software-testing> [Basic of Software Testing]
2. <https://www.guru99.com/functional-testing.html> [Functional Testing]
3. <http://www.softwaretestinggenius.com/download/bgstpadmini.pdf> [Software Testing Life Cycle]

Course Outcomes: Upon successful completion of the course, the students will :

C01 : Understand the role of testing in software development.

C02 : Apply the test case design techniques.

C03 : Acquire the various levels of testing.

C04 : Document test plan and design.

C05 : Execute test case and measure the results.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Basic concepts of Software Testing	✓				
2	White Box Testing Techniques	✓	✓			
3	Black Box Testing Techniques	✓	✓			
4	Levels of Testing	✓		✓		
5	Test Planning and Design	✓			✓	
6	Test Execution and Measurements	✓				✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	1	1	1	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	2	2	3	3	2	2	2	2	2	1	1	1	3	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3	3	2
CO5	3	3	3	3	3	3	3	3	3	2	3	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**MS219: Object Oriented Programming using Java
(200 Marks)**

Contact Hours: 06

Pre-requisite: None.

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for Object Oriented Programming concepts will be discussed. Emphasize will be given on Core Java concepts, various Java packages, Exception handling, Multithreading and The collection framework. During Practical sessions, students will focus on developing simple Java programs to understand the concepts practically.

Outline of the Course:

Unit No.	Title of the Unit	Minimum No. of Hours	
		Theory	Practical
1	Object Oriented Programming and Introduction to Java	06	36
2	Fundamental programming structures in java	07	
3	Understanding class, objects, methods and constructors	07	
4	Inheritance, Abstract class and Interface	05	
5	Java Package concept and Exception	05	
6	Multithreading and Collection Framework	06	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit – I: Object Oriented Programming and Introduction to Java Hours: 06

Programming Paradigms (Procedure Oriented and Object Oriented), Object Oriented

Programming concepts, comparison of POP and OOP

Introduction to Java programming language, History of Java, Features of Java, Structure of Java program, creating an application in Java, Compiling and executing applications in java, command line arguments

Unit – II: Fundamental programming structures in java Hours: 07

Data types, Variables, Literals, Arrays Operators:

Arithmetic Operators, Basic Assignment Operators, Relational Operators, Boolean Logical Operators, Ternary Operator, Operator Precedence Control Statements: Java's Selection Statements, Switch, Nested Switch, Iteration Constructs, Continue, Return. Control Statements:

Selection: If statements, Switch statement

Iteration: loop statements, for each loop

Break Statements: break, continue and return

Unit – III: Understanding class, objects, methods and constructors Hours: 07

Defining a class in Java, creating objects, Instance variables, Initializer blocks, methods and its overloading, The String class and its useful methods, Constructors and its overloading, this keyword, Objects as Parameters, returning objects, static keyword, Garbage collection and finalize method

Unit – IV: Inheritance, Abstract class and Interface Hours: 05

Inheritance and its types, method overriding, final keyword, super keyword, Type casting, Abstract classes and abstract methods, Interfaces

Unit – V: Java Package concept and Exception Handling Hours: 05

Defining a Package, finding packages and CLASSPATH, Important Packages, Access protection, importing packages, User defined packages. Wrapper Classes, Boxing and Unboxing conversions

Fundamentals of Exception Handling, Types of Exceptions, keywords used in Exception handling, Multiple catch clauses, Nested try Statements, Java Built in Exceptions, User Defined Exceptions

Unit – VI: Multithreading and Collection Framework Hours: 06 Multithreaded

Programming:

The Java Thread Model, Priorities, Messaging, The Thread class and the Runnable Interface, creating a thread, Creating Multiple Threads, Daemon thread, Synchronization and Deadlock, The volatile modifier.

The classes and interfaces for the collection framework - Collection, Set, List, Map, ArrayList, Vector, Hashtable and HashMap, Generics in Collection framework.

The Utility Classes: The Enumeration Interface, StringTokenizer, Date, Calendar and Date Comparison

Core Books:

1. Pravin M. Jain: The class of JAVA, Pearson Education, 2011.
2. Herbert Schildt: The Complete Reference Java J2SE, 5th Edition, TMH Publishing Company Ltd, NewDelhi.

Reference Books:

1. Ivor Horton: Beginning Java JDK, 5th Edition, Wiley Press, 2008.
2. Cay Horstmann and Gary Cornell: Core Java Volume 1, 8th Edition or later, Pearson Education, 2008.
3. Ken Arnold, James Gosling and David Holmes: The Java Programming Language, 4th Edition, Addison Wesley.

Web References:

1. <http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-092introduction-to-programming-in-java-january-iap-2010/lecture-notes/> [Lecture Notes in PDF]
2. <http://www.javatpoint.com/java-tutorial> [Notes]

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Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Understanding object oriented concepts and its implementation through Java programming language
- C02 : Able to distinguish between Traditional/Procedure Oriented Programming and OOP.
- C03 : Be familiar with tools, techniques and IDE for developing java application.
- C04 : Be familiar with the concepts of packages and utilizing inbuilt classes from java packages.
- C05 : Be able to develop console java programs and applications.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Object Oriented Programming and Introduction to Java	✓	✓			✓
2	Fundamental programming structures in java		✓			✓
3	Understanding class, objects, methods and constructors	✓		✓		✓
4	Inheritance, Abstract class and Interface	✓				✓
5	Java Package concept and Exception			✓	✓	✓
6	Multithreading and Collection Framework				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	2	2	2	1	-	-	-	-	-	-	-	-	2
CO2	3	2	2	3	2	2	-	3	-	-	-	-	3	2
CO3	-	-	3	3	2	-	-	-	-	-	-	-	-	-
CO4	3	2	2	-	2	2	-	3	-	-	-	-	3	2
CO5	2	-	-	-	2	-	3	3	-	-	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**MS220 : HTTP Web Service for Enterprise Application
(200 Marks)**

Contact Hours: 06

Pre-requisite: Knowledge of C#,ASP.NET MVC, HTML, CSS, and have some understanding of JavaScript. Database connectivity knowledge.

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for advanced features of WEB API with ASP.NET will be given. During Practical sessions, students will be required to develop Web API using concepts discussed during class.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Overview of ASP.NET Web API	07	36
2	Web API Controller and Model	06	
3	Web API Routing	05	
4	Web API Formatter and Exception Handling	05	
5	Hosting of Web API	05	
6	Managing Database and Consume Web API	08	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit - I: Overview of ASP.NET WEB API Hours:07

Introduction to Service Oriented Architecture, Introduction to WEB API, Characteristics of ASP.NET WEB API, versions of Web API, Difference among web service, Window Communication Foundation and WEB API, Test Web API Fiddler and Postman

Unit - II: Web API Controller and Model Hours:06

Use of Web Api Controller in Controller class of Web API, Characteristics of Web API Controller, Difference between Web API and MVC controller, Action Method Naming Conventions, Action Result: void, HttpResponseMessage, IHttpActionResult and Other types, Use of Model in Web API, Validation in Model

Unit - III: Web API Routing Hours:05

Routing in Web API, Types of routing: Convention-based Routing and Attribute Routing, Routing and Action Selection and Attribute Routing

Unit - IV: Web API Formatter and Exception Handling Hours:05

Data Formatter, Media Type Formatter, Web API Filter, Filter of Exception, HttpResponseMessage, Exception Filters, Registering Exception Filters, HttpError

Unit –V: Hosting of Web API and Securing Web API Hours:05

IIS Hosting, Self-Hosting, hosting on the Open Web Interface for .NET, Hosting for Azure Mobile Services, Encrypt-Decrypt Text, Authentication: Filter, Basic, Forms and Integrate Window Authentication

Unit - VI: Managing Database and Consume Web API Hours:08

CRUD operation with Entity Framework in Web API, Consume Web API through GET, POST, PUT and DELETE

Core Books:

1. Phillip Wojcieszyn: “ASP .NET WEB API 2 Recipes”, Apress, April 2011.
2. Uurlu, Ali, Zeitler, Alexander, Kheyrollahi, Ali: “Pro ASP.NET Web API HTTP Web Services in ASP.NET”, Apress, 2013.

Reference Books:

1. Akhil Mittal: “Diving into ASP.NET Web API”, 2016.
2. Ricardo Peres:” Entity Framework Core Cookbook”, 2nd Edition, Kindle Edition
3. Shankar Kambhampaty, “Service-Oriented Architecture: For Enterprise Applications”, WILEY India Edition

Web References:

1. <http://www.tutorialsteacher.com/webapi/web-api-tutorials>
2. <https://www.udemy.com/learning-aspnet-web-api/>

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Be familiar with SOA and technology to implement SOA in ASP.NET
- C02 : Be aware about use of Model and Controller in Web API
- C03 : Be familiar with Routing concept in Web API
- C04 : Be able to use different formatter and exception handling in Web API
- C05 : Be able to Host Web API using various ways
- C06: Be able to manage database connectivity thorough Entity Framework
- C07: Be able to consume Web API through different method of HTTP

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Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes						
		C01	C02	C03	C04	C05	C06	C07
1	Overview of ASP.NET WEB API	√						
2	Web API Controller and Model		√					
3	Web API Routing			√				
4	Web API Formatter and Exception Handling				√			
5	Hosting of Web API					√		
6	Managing Database and Consume Web API						√	√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	2	-	1	-	-	-	1	2	3	-
CO2	2	1	3	2	3	-	1	-	-	-	-	1	2	-
CO3	-	2	2	1	-	-	1	-	-	-	-	-	1	-
CO4	1	3	3	2	-	-	2	-	-	-	-	1	2	-
CO5	-	-	1	-	1	-	1	2	-	-	1	-	2	-
CO6	-	1	-	3	2	-	-	-	-	-	-	-	1	-
CO7	-	-	-	-	2	-	2	1	-	-	1	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS221 : Multi Paradigm Scripting using Python - II
(Marks 200)

Contact Hours: 06

Pre-requisite: Object Oriented Programming Using C++/JAVA, MPPL – I.

Methodology & Pedagogy: Theory sessions are required to address computational power of python through framework like Django. Practical sessions demonstrate the implementation of the concepts which are taught during the theory sessions. Case study will help the students to come out with one working module in any of the paradigm through python. Practical session should be full of live application demonstrations and integration of service oriented architecture also.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	
1	Introduction to Django	05	36
2	Getting started with Django	05	
3	Django URL Patterns and Views	08	
4	Django Forms, Django & REST APIs	06	
5	Unit Testing with Django	06	
6	Admin, Security, Performance & Optimization	06	

Total Hours (Theory): 36

Total Hours(Lab): 36

Total Hours: 72

Detailed Syllabus:

Unit – I: Introduction to Django Hours: 05

What is Django? Python in context of Django. History of Django. MVC Feature of Django. Model, View and Template. DRY Programming using Django. Installation guide for Django. Special Features of Django.

Unit – II: Getting started with Django Hours: 05

Installing Multi and Django. Setting up database. Django with PostgreSQL, SQLite3, MySQL. Django without Database. Django Project with Development Server. Basics of Dynamic Web Pages. Overview of Django Template System.

Unit – III: Django URL Patterns and Views Hours: 08

Designing a good URL scheme. URLconfTricks, Streamlining Function, Imports, Using Multiple View Prefixes, Special-Casing URLs in Debug Mode, Using Named Groups, Understanding the Matching/Grouping Algorithm, Passing Extra Options to View Functions Using Default View Arguments, Special-Casing Views, Capturing Text in URLs, Determining What the URLconf Searches Against, Including Other URLconfs, How Captured Parameters Work with include (), How Extra URLconf Options Work with include().

Generic Views, Usage of Generic View, Generic View Object. Extending Generic View.

Unit – IV: Django Forms, Django & REST APIs: Hours: 06

Form classes, Validation, Authentication, Advanced Forms processing techniques: Search forms, Creating Feedback form, Submission process of the form, Custom validation rules, Custom look and feel, Creating forms for models. REST APIs and its implementation. Django REST framework, DjangoPiston

Unit – V: Unit Testing with Django: Hours: 06

Overview / Refresher on Unit Testing and why it's good, Using Python's unittest2 library, Test, Test Databases, Doctests, Debugging Best Practices

Unit – VI: Admin, Security, Performance & Optimization: Hours: 06

Admin characterization in Django, Security facility in Django, Performance issues Django and Optimization in Django framework.

Core Books:

1. Adrian Holovaty, Jacob K. Moss: The Definitive Guide to Django:Web Development Done Right
2. Jeff Forcier, Paul Bissex, Wesley Chun: Python Web Development with Django, Pearson Education,2008.
3. Dusty Philips: Python 3 Object oriented Programming , PACKT publishing, 2010.

Reference Books:

1. Adrian Holovaty, Jacob Kaplan-Moss, et al.: The Django Book, 2013
2. Nathan Yergler: Effective Django, 2015
3. Steve Holden: Python Web Programming, 1st edition,2002.

Web References:

1. <https://www.djangoproject.com/start/> [For Django Projects]
2. <https://djangoforbeginners.com/initial-setup/> [For Django Setups]
3. <https://docs.djangoproject.com/en/2.1/intro/tutorial01/> [For Django app]

Course Outcomes: Here's what you will learn!

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- C01 : Understand Django fundamentals and use its concepts to build and deploy robust web applications and apps.
- C02: Learn how to configure Django for powerful databases and create the Django admin interface
- C03 : Learn about Django URL patterns and views and deploy Django applications.
- C04 : Learn about Django's security implications and how to create safe web applications with it.
- C05 : To get some idea about unit testing concepts and its implementation through Django.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Django	✓				
2	Getting started with Django	✓	✓			
3	Django URL Patterns and Views		✓			
4	Django Forms, Django & REST APIs		✓			✓
5	Unit Testing with Django			✓		✓
6	Admin, Security, Performance & Optimization				✓	

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	2	2	2	1	-	-	-	-	-	-	-	3	3
CO2	3	2	2	3	2	2	-	3	-	-	-	-	2	3
CO3	-	-	3	3	2	-	-	-	-	-	-	-	2	3
CO4	3	2	2	-	2	2	-	3	-	-	-	-	3	3
CO5	2	-	-	-	2	-	3	3	-	-	-	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS222 : Game Development - I
(Marks 200)

Contact Hours: 06

Pre-requisite: Knowledge of C++, physics and mathematics concepts.

Methodology & Pedagogy: During theory lectures illustrations Graphics, animation and various concepts regarding Game Development. Emphasize will be given on some mathematical and physics concepts, Fundamental of Graphics and Graphics Programming, Vectors, Image Data, 2D graphics, Collision detections, Fundamental of Game programming, Gameloop, Game Engine and many more. During Practical sessions, students will be develop Simple 2D graphics, Implement animation on that graphics, Handles Input and Output of data according to event generation. Student will develop simple 2D game using Visual C++.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Introduction to Graphics and Game	04	36
2	Essential Mathematics and Physics Concepts	07	
3	Fundamentals of Graphics Programming	05	
4	Understanding Animation, Transformations and Collision detection	07	
5	Fundamentals of Game Programming	09	
6	Basics of Game Engine	04	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit – I: Introduction to Graphics and Game Hours: 04

The World of Computer Graphics, Current and Future Application Areas, User-Interface Considerations, History of Games and Game Development

Unit – II: Essential Mathematics and Physics Concepts Hours: 07

Vector Addition, Subtraction, Dot Product, Cross Product, Magnitude, Finding the Normal of a Vector, Projection of one vector over another vector, Matrix and operations: Matrix Addition, Subtraction, Multiplication, Transpose, and Inverse of a Matrix, Basic Trigonometry: Meaning of Sin, Cosine, and Tan functions, Cartesian coordinate system, Polar Coordinate System, Nested Coordinate systems

Physics Concepts: Speed, Velocity, Acceleration, Mass, Force, Gravity, Friction

Unit – III: Fundamentals of Graphics Programming Hours: 05

Video Ram, Pixels and screen sizes , Graphics Card , Frame Buffer , Double Buffer , Screen ,Refresh Rates (on LCD display units) , Screen Resolutions , Vector Graphics and Raster Graphics , Image Data - How are images represented in a Bitmap file, How are images drawn on a screen

Unit – IV: Understanding Animation, Transformations and Collision detection Hours: 07

Programming Simple Animations and understanding animation frame rates, Concept of Sprites in Animation, Line Drawing Algorithm, Circle Drawing Algorithm, Transformations - Translate, Rotate, Scale, Transformation Matrix, Collision detection techniques for 2D Shapes: Circle to Circle, Rectangle to Rectangle, Circle to Line, Circle to Rectangle, Line to Line

Unit – V: Fundamentals of Game Programming Hours: 09

Application and Window Management , Input Management , File Input and Output , Audio, Programming , Game loop , Frame rate independent entity behaviors

Unit – VI: Basics of Game Engine Hours: 05

Introduction to Game Engine, Types of Game Engines, Some of the most popular Game Engines of today: Unity, Unreal, Cocos2d-X, Box2d

Core Books:

1. Laszlo Michael , Computational Geometry and Computer Graphics in C++, Pearson Education; First edition, 2017
2. Roger Mayne, Introduction to Windows and Graphics Programming with Visual C++.NET, World Scientific Publishing Co. Pte. Ltd., 2005
3. John Horton, Beginning C++ Game Programming, Packt Publishing Limited, 2016
4. Michael Dawson , Beginning C++ Through Game Programming, Cengage; 4th edition, 2014
5. Eric Lengyel, Foundations of Game Engine Development, Volume 1: Mathematics, Terathon Software LLC; 1 edition, 2017

Reference Books:

1. John F. Hughes , Andries van Dam , Morgan McGuire , David F. Sklar , James D. Foley , Steven K. Feiner , Kurt Akeley , "Computer Graphics: Principles and Practice in C", 3rd Edition, AddisonWesley Professional, 2013
2. David Conger, Ron Little, Creating Games in C++: A Step-by-step Guide, Pearson Education, 2006
3. By David Conger, Ron Little
4. Fabien Sanglard ,Game Engine Black Book: Wolfenstein 3D, CreateSpace Independent Publishing Platform; 1 edition
5. Constantine Pozrikidis, Introduction to C++ Programming and Graphics, Springer-Verlag New York Inc.; 2007
6. James M. Van Verth, Lars M. Bishop, Essential Mathematics for Games and Interactive Applications, A K Peters/CRC Press; 3 edition, 2015

Web References:

1. <http://cplusplus.happycodings.com/computer-graphics/> [Graphics Tutorial]
2. <http://cplusplus.happycodings.com/games/> [Game Tutorial]
3. <https://www.geeksforgeeks.org/basic-graphic-programming-in-c/>
4. <https://www.thecrazyprogrammer.com/2013/04/simple-program-to-create-moving-car-in.html>
5. <http://mytechnotrick.blogspot.com/2015/07/#>
6. <https://www.youtube.com/watch?v=hfBR6ULOOjw>

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Be having basic knowledge of computer graphics
- C02 : Be familiar with tools and techniques used for graphics and game programming
- C03 : Be able to perform mathematic calculation required for game programming
- C04 : Be familiar with the various image data and shapes.
- C05 : Be able to create animated and interactive graphics which lead to develop a game.

Course Outcomes Mapping:

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Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Graphics and Game	✓				✓
2	Essential Mathematics and Physics Concepts		✓			✓
3	Fundamentals of Graphics Programming	✓	✓	✓	✓	✓
4	Understanding Animation, Transformations and Collision detection	✓			✓	✓
5	Fundamentals of Game Programming	✓		✓	✓	✓
6	Basics of Game Engine		✓			✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	2	-	1	-	-	-	1	2	3	-
CO2	2	1	3	2	3	-	1	-	-	-	-	1	2	-
CO3	-	2	2	1	-	-	1	-	-	-	-	-	1	-
CO4	1	3	3	2	-	-	2	-	-	-	-	1	2	-
CO5	-	-	1	-	1	-	1	2	-	-	1	-	2	-
CO6	-	1	-	3	2	-	-	-	-	-	-	-	1	-
CO7	-	-	-	-	2	-	2	1	-	-	1	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS106.02 C: ACADEMIC WRITING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	2
Marks	--	100	--	100	

Pre-requisite courses:

- An Intermediate Guide to Writing in English for University Study
<https://www.futurelearn.com/courses/english-for-study-intermediate/4/todo/62943>

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Academic Writing and Research Process	05
2.	Anatomy of Academic Writing	05
3.	Key Academic Skills	05
4.	Accuracy in Academic Writing	05
5.	Using and Citing Sources of Ideas	05
6.	Contemporary Practices in Academic Writing	05
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

1.	Academic Writing and Research Process	5 Hours	
	Introduction to Academic Writing, Academic Writing as a Part of Research, Types of Academic Writing, Features of Academic Writing, Importance of Good Academic Writing in various Academic Works		
2.	Anatomy of Academic Writing	5 Hours	
	Academic Vocabulary, Simple and Complex Sentences, Organizing Paragraphs, The Writing Process, Adopting Academic Writing Style		
3.	Key Academic Skills	5 Hours	
	Note – taking, Note – making, Paraphrasing, Summarizing		

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4.	Accuracy in Academic Writing	5 Hours	
	Lexical Range, Academic Language and Structures, Elements of Writing, Proof Reading, Editing, and Rewriting		
5.	Using and Citing Sources of Ideas	5 Hours	
	Academic Texts and their Types, Intellectual Honesty in Academic Writing, Avoiding Plagiarism – Idea Theft, Degrees of Plagiarism, Types of Borrowing, Anatomy of Citations, Common Citation Styles		
6.	Contemporary Practices in Academic Writing	5 Hours	
	Analytical Essays, Graph / Table / Process Interpretation and Description, Writing Reports and Abstract, Writing Research / Concept Papers		

Course Outcome (COs):

At the end of the course, the students

CO1	Will have sound understanding of the concept and applications of academic writing
CO2	Will have acquired enough knowledge of academic writing style, strategy and approach
CO3	Will be able to demonstrate error free and effective academic writing
CO4	Will be able to demonstrate ability to work on project/report/paper writing
CO5	Will have the sound understanding of the Research and Research Methodology
CO6	Will be effectively communicating in diverse academic and professional settings.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO5	-	-	-	-	-	-	3	-	3	-	-	-	-	2
CO6	-	-	-	-	-	-	2	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

- **Text book:**

1. Academic Writing for International Students, Routledge
2. Academic Writing: A Guide for Management Students and Researchers. Monipally, M.M. & Pawar, B.S. Sage. 2010. New Delhi
3. Effective Academic Writing Level - 1,2,3,4 (Second Edition) By: Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; Publisher: Oxford

- **Reference book:**

1. Writing Your Thesis (2nd Edition) by Paul Oliver, Sage
2. Development Communication In Practice by Vilanilam V J, Sage
3. Intercultural Communication by Mingsheng Li, Patel Fay, Sage

- **Web material:**

www.owl.perdue.edu

TEACHING SCHEME & DETAILED SYLLABUS

FOR

**M.Sc. (IT) PROGRAMME
(3rd SEMESTER)**

EFFECTIVE FROM ACADEMIC YEAR 2021-22

**Teaching and Examination Scheme (M.Sc(IT) Programme)
Effective from Year 2021-22**

SEMESTER-III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Pract	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
MS320	Open Source Frameworks	3	3	6	6	10	20	70	15	15	70	200
MS321	Advanced Java	3	3	6	6	10	20	70	15	15	70	200
MS322	JavaScript Frameworks	-	3	3	3	-	-	-	15	15	70	100
MS323	Mobile Application technology	3	3	6	6	10	20	70	15	15	70	200
MS324-MS327	Elective-III	3	3	6	6	10	20	70	15	15	70	200
CL144.03C	Environmental Sciences	-	2	2	2	-		-	30		70	100
		12	17	29	29	400			600			1000

Elective-III

1. MS324 Game Development-II
2. MS325 Data Analytics
3. MS326 Ruby on rails
4. MS327 Internetworking with TCP/IP

MS320 : Open Source Frameworks
(200 Marks)

Contact Hours: 06

Pre-requisite: MS216: Web Development using Open Source Technology

Methodology & Pedagogy: In theory sessions, the emphasis will be given on Introduction to Object Oriented Programming in PHP, Introduction, installation and configuration of laravel framework, routing and artisan in laravel, working with blade template and SQL interaction for database transaction and object relational mapper which is eloquent ORM. During practical sessions, students shall implement the concepts, which are taught during theory sessions.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Object Oriented Programming in PHP	05	36
2	Bootstrap Basics	05	
3	Introduction to Laravel with Installation & Configuration	06	
4	Routing in Laravel & Introduction to Artisan	08	
5	Blade Template and SQL Interaction	06	
6	Eloquent ORM	06	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit – I: Object Oriented Programming in PHP Hours: 05

The Basics, Properties, Class Constants, Auto loading Classes, Constructors and Destructors, Visibility, Object Inheritance, Scope Resolution Operator (::), Static Keyword, Class Abstraction, Object Interfaces, Anonymous classes, Overloading, Object Iteration, Magic Methods, Final Keyword, Object Cloning, Comparing Objects, Type Hinting, Late Static Bindings, Objects and references

Unit – II: Bootstrap Basics Hours: 05

Introduction: File Structure, Basic HTML Template, Global Styles, Default Grid System, Basic Grid HTML, Offsetting Columns, Nesting Columns, Fluid Grid System, Container Layouts, Responsive Design, Introduction to Responsive Design.

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Implementation: Typography, Code, Tables, Forms, Buttons, Images, Icons, Glyphicons, ropdwn Menus, Button Groups, Button with Dropdowns, Navigations, Navbar, Breadcrumb, Pagination, label, badges, Typographic elements, thumbnails, alerts, progress bar, wells.

Unit – III: Introduction to Laravel with Installation & Configuration Hours: 06

What is Laravel, features, MVC architecture, structure of laravel application (laravel directory structure),

Basic requirements for Laravel, Using Laravel Installer, Using Composer, Linux & Windows, Finding and installing new packages.

Introduction, Environment configuration, Protecting sensitive configuration, Maintenance mode, database configuration (setting database connection parameter for laravel and artisan)

Unit - IV: Routing in Laravel & Introduction to Artisan Hours: 08

Basic Routing, Route Parameters, Route Filters, Named Routes, Route Groups, Sub-Domain Routing, Route Prefixing, Route, Model Binding, Throwing 404 Errors, Routing to Controllers

Artisan Command Line Tool, database creation, artisan migration, migration structure, creation migration, Database seeding

Unit – V: Blade Template and SQL Interaction Hours: 06

Template inheritance, Master layout, Extending the master layout, display variables, Blade conditional statements, Blade Loops, Executing PHP functions in blade

Introduction, Running Raw SQL Queries, Database Transactions

Unit – VI: Eloquent ORM Hours: 06

Eloquent ORM Models: Naming conventions, table name & primary keys, timestamps

Basic Operations: Create, Retrieve, Update, Delete Using Models, displaying data from models in views.

Core Books:

1. Jake Spurlock : Bootstrap, O'reilly, May 2013.
2. Martin Bean: Laravel 5 Essentials, Packet Publishing, April 2015.

Reference Books:

1. Sheikh Heera: Laravel Reference Guide, Packet Publication, June 2016.
2. Matt Stauffer, Laravel: Up and Running: A Framework for Building Modern PHP Apps, paperback, 2016.

Web References:

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1. https://www.tutorialspoint.com/php/php_object_oriented.htm [Object-Oriented Programming in PHP]
2. <https://laravel.com/docs/5.2> [Online Laravel 5.2 Documentation]
3. <https://code.tutsplus.com/tutorials/build-web-apps-from-scratch-with-laravel-the-eloquent-orm-net-25631> [For Eloquent ORM]
4. <https://laravel.com/docs/5.7> [For Blade Template, Routing, Database Connectivity]

Course Outcomes: Upon successful completion of the course, the students will:

- CO1 : Be able to understand object oriented concepts in PHP.
- CO2 : Understand the basics of bootstrap to create responsive web application.
- CO3 : Be able to configure and install the Laravel framework.
- CO4 : Able to create MVC based application using the Laravel framework.
- CO5: Understand Template integration and database communication in the web application development. Also able to interact with command-line interface included with Laravel.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Object Oriented Programming in PHP	✓			✓	
2	Bootstrap Basics		✓		✓	✓
3	Introduction to Laravel with Installation & Configuration			✓	✓	
4	Routing in Laravel & Introduction to Artisan				✓	✓
5	Blade Template and SQL Interaction				✓	✓
6	Eloquent ORM				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	1	-	-	3	-	-	-	-	1	3	-
CO2	1	-	-	1	1	-	3	-	-	-	-	1	3	-
CO3	-	3	2	1	2	-	3	1	-	-	-	1	3	-
CO4	2	2	3	2	2	-	3	2	-	-	1	2	3	-
CO5	2	1	2	2	2	-	3	3	-	-	1	2	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**MS321 : Advanced Java
(200 Marks)**

Contact Hours: 06

Pre-requisite: MS218 - Object Oriented Programming using Java

Methodology & Pedagogy: During theory sessions topics related to technologies and techniques used in the development of Web-based applications. During Practical sessions, students will be required to develop Applications using the concepts of JDBC, JSP, JSTL and Servlets. Student will also be explored to MVC architecture.

Outline of the Course:

Unit No.	Title of the Unit	Minimum No. of Hours	
		Theory	Practical
1	Database Programming	07	36
2	Java Web Application Components	05	
3	Java Server Pages	07	
4	JSP Standard Tag Library	04	
5	Working with Servlets	08	
6	Advanced Servlet Features & Security	05	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit – I: Database Programming Hours: 07

The Design of JDBC, The Structured Query Language, JDBC Configuration, Executing SQL Statements, Query Execution, Scrollable and Updatable Result Sets, Row Sets, Metadata, Transactions

Unit – II: Java Web Application Components Hours: 05

Understanding Web Applications: Understanding Web Components, Servlets (Overview), JSP Pages, Introducing the MVC Design Pattern, Understanding Containers, Packaging Web Applications, Web Application Structure, JAR Files, WAR Files, HTTP, GET Request Method, POST Request Method, GET & POST in HTML Form Processing, Other Request Methods, The HTTP Response, Using Deployment Descriptors

Unit – III: Java Server Pages Hours: 07

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Introducing JSP Technology – Understanding the Page Life Cycle, JSP Documents, JSP Document Syntax, A Simple JSP Document, HTTP, HttpJspPage Interface.

Introducing JSP Syntax and Usage – General Rules of Syntax, JSP Elements, Available Object Scope, Implicit Objects, Scripting Elements, Comments, JSP Expression Language

Unit – IV: JSP Standard Tag Library Hours: 04

Introduction to JSTL, Core Tag Library

Unit – V: Working with Servlets Hours: 08

Working with Servlets (annotation based): Introducing Servlet, Introducing Servlet & the MVC Pattern, Introducing javax.servlet Package, Introducing the Servlet Interface, Introducing the GenericServlet Class, Introducing HTTP & Servlets, Understanding the Request/ Response Cycle, Input & Output Streams, Introducing Servlet/ Container Communication, Introducing ServletContext, Understanding the Deployment Descriptor, Introducing ServletContext Lifecycle Classes, RequestDispatcher Interface, Using Filters & RequestDispatcher

Unit – VI: Advanced Servlet Features & Security Hours: 05

Understanding the Stateless nature of HTTP, Why Track Client Identity & State? Maintain Sessions, Session Management Using the Servlet API, and Concepts of Filters.

Core Books:

1. Joel Murach, Michael Urban: Java Servlet & JSP, 3rd Edition, Murach's Production.
2. Budi Kurniawan, Servlet & JSP A Tutorial, 2nd Edition, Braine Software.
3. Marty Hall, Larry Brown: Core Servlets and JavaServer Pages, Volume 2, Advanced Technologies, 2nd Edition, Pearson Education, 2008.

Reference Books:

1. Marty Hall, Larry Brown: Core Servlets and JavaServer Pages, Volume 1, Core Technologies, 2nd Edition, Pearson Education, 2003.
2. Bryan Basham, Kathy Sierra and Bert Bates: Head First Servlet and JSP, O'Reilly Publication, 1st Edition.

Web References:

1. <http://courses.coreservlets.com/Course-Materials/csajsp2.html> [Servlet Basics]
2. <http://www.ceit.es/assignaturas/InteInfo/Recursos/Servlets/JavaServlets.pdf> [Servlet Tutorial PDF]
3. <http://www.msuniv.ac.in/AdvancedJavaProgrammingwithDatabaseApplication.pdf> [JDBC Tutorial]

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Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Be familiar with Enterprise Edition Framework of Java
- C02 : Be familiar with web application development using Servlet
- C03 : Be familiar with web application development using JSP
- C04 : Able to create web application using advance Java
- C05 : Able to develop MVC application using Java enterprise edition

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Database Programming				✓	✓
2	Java Web Application Components	✓	✓	✓	✓	✓
3	Java Server Pages	✓	✓	✓		
4	JSP Standard Tag Library	✓		✓	✓	✓
5	Working with Servlets	✓	✓		✓	✓
6	Advanced Servlet Features & Security	✓			✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3	2	2	-	-	2	1	-	1	-	-	-	1
CO2	2	2	2	2	-	-	2	1	-	1	-	-	-	1
CO3	2	2	2	2	-	-	2	1	-	1	-	-	-	1
CO4	-	2	3	-	3	-	2	3	-	1	2	2	3	1
CO5	-	2	3	-	3	-	2	3	-	1	2	2	3	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**MS322 : Javascript Frameworks
(100 Marks)**

Contact Hours: 03

Pre-requisite: Basic knowledge of HTML5, CSS3.

Methodology & Pedagogy: During Practical sessions, students will be required to design and develop responsive web sites using HTML5, CSS3, AngularJS and NodeJS.

Outline of the Course:

Week No.	Content
1-2	• Introduction to JavaScript. • The Java Document Object Model (DOM): Introduction, Objects in HTML, Object hierarchy, Event handling.
3-4	• Introduction to AngularJS. • Creating basic AngularJS application using its component, How AngularJS Integrates with HTML. • Working with AngularJS Expressions □ Data Binding in AngularJS. • Working with AngularJS Directives. • Working with the variables. • Creating a loop in AngularJS. • Handling Events
5-7	• Working with Controller. • MVVM design Architecture with AngularJS. • Working with AngularJS Module. • Working with forms and its elements.
8-10	• Working with AngularJS HTML DOM • Working with Database using AngularJS.
11-12	• Fundamentals of Node JS, Installation of Node JS • Basics of NodeJS module. • Node JS Events. • Developing Model-View-Controller layer.

Total Hours :36

Core Books:

1. Jeffry Houser : “Learn With: Angular 5, Bootstrap, and NodeJS”, Kindle Edition,
2. Shyam Seshadri Brad Green: “AngularJS – Up and Running, Brad Green”, Second Edition, O'REILLY
3. Basarat Ali Syed: “Beginning Node.js”, Apress Publication.

Reference Books:

1. Agus Kurniawan: "AngularJS Programming by Example 2017 Edition", Kindle Edition.
2. Adam Freeman: " Pro AngularJS 2017 Edition", Apress.
3. Krasimir Tsonev: "Node.js By Example", Packt Publishing

Web References:

1. <http://www.w3schools.com/angular/default.asp> [Tutorial link for AngularJS]
2. <http://www.tutorialspoint.com/angularjs/> [Tutorial link for AngularJS]
3. https://www.tutorialspoint.com/angularjs/angularjs_tutorial.pdf [E-book for AngularJS]
4. <http://www.tutorialsteacher.com/nodejs/nodejs-modules> [Tutorial link for NodeJS]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Be able to understand Javascript basics.
- C02 : Be familiar with JavaScript Framework (AngularJS and NodeJS) and its applications.
- C03 : Be able to create a single page application using AngularJS.
- C04 : Create web application using the MVVM pattern with JavaScript Framework. Also, able to maintain the two-way data binding between model and view.
- C05 : Be able to create responsive web page with database using AngularJS.

Course Outcomes Mapping:

Lab	Content	Course Outcomes				
		C01	C02	C03	C04	C05
1-2	Introduction to JavaScript Framework.	✓	✓			
3-4	Basics of AngularJS .		✓	✓		
5-7	Working With MVVM Architecture.			✓	✓	
8-10	Working with AngularJS HTML DOM & Database.			✓	✓	✓
11-12	Basic Fundamental of NodeJS		✓			

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Course Articulation Matrix:

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3	2	3	2	2	1	1	-	3	-
CO2	3	3	3	2	3	2	3	2	2	2	2	2	3	2
CO3	3	3	3	2	3	2	3	2	2	2	2	-	3	-
CO4	3	3	3	3	3	3	3	3	2	2	3	2	3	2
CO5	3	3	3	3	3	3	3	3	3	2	3	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**MS323 - Mobile Application Technology
(200 Marks)**

Contact Hours: 06

Pre-requisite: MS218 - Object Oriented Programming using Java.

Instructional Methods & Pedagogy: During theory lectures illustrations emphasizing the usage of Android- the Mobile Application Development platform will be given. During Practical sessions, students will be required to develop Mobile Application in Android. Student shall also develop applications with elegant user interface that deal with data storage and Web Service.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Introduction Android and Android Development tools	7	36
2	User Interface I	8	
3	User Interface II	5	
4	Working with files and Shared Preferences	5	
5	Working with Database	6	
6	Web Service Integration	5	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit - I: Introduction Android and Android Development Tool

Hours: 07

Introduction to Mobile Applications and Android & basics of Kotlin language, Setting up development environment, Dalvik Virtual Machine, Android Application Structure, DDMS, Android Manifest File, Gradle, Android permissions, APK ,Basic Building blocks – Activities, Services, Broadcast Receivers & Content providers ,Basic Views: Toast, TextView, EditText, Button , Components for communication - Intents & Intent Filters : Intents: Explicit Intents, Implicit Intents, Switching between activities and passing data between activities using Intents. Logging

Unit - II: User Interface – I

Hours: 08

Working with Containers: LinerLayout, RelativeLayout, TableLayout, ScrollView, and FrameLayout, Fragment Views: ImageButton, CheckBox, ToggleButton, RadioButton, RadioGroup, ProgressBar, AutoComplete TextView Picker Views: TimePicker and DatePicker views Adapters: ArrayAdapter, BaseAdapter ,ListView and ListActivity, GridView, Gallery

Unit - III: User Interface – II Hours: 08

Menus: Option menu, Context menu, Sub menu, Tabs and TabActivity ,Alert Dialogs,Navigation Drawer, Android Toolbar

Unit - IV: Working with Files, Shared Preference, Animation Hour: 07

Basics of Storage in Android, Shared Preference : Save key-value data, Save files on Device,Sharing of data, Animation

Unit - V: Working with Database Hour: 05

Basics of SQLite Database, CRUD Operations with Android

Unit – VI: Web Services Integration Hour: 04

Web service integration with Android, Deploying Web Service

Core Books:

1. Wei-Meng Lee: Beginning Android 4 Application Development, Wiley India Pvt Ltd.
2. Reto Meier: Professional Android 2 Application Development, Wrox publication
3. Mark L. Murphy: The Busy Coder's Guide to Android Development

Reference Books:

1. Dawn Griffiths, David Griffiths: Head First Android Development, O'REILLY publication
2. Mark L. Murphy: Beginning Android 2, APRESS publication
3. Mark L. Murphy : The Busy Coder's Guide to Advanced Android Development, Commons Ware, LLC

Web References:

1. <https://www.tutorialspoint.com/android/> [Android Tutorials]
2. <https://developer.android.com/guide/> [Android Tutorials]
3. www.vogella.com/tutorials/android.html [Android Practical Tutorials]
4. <https://www.androidhive.info/> [Android Practical Tutorials]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Be familiar with Android OS
- C02 : Be familiar with Android Development Tools
- C03 : Be able to create proper look and feel of mobile application
- C04 : Be able to create interactive mobile application that handle data
- C05 : Be able to create mobile application that works web service.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction Android and Android Development tools	✓	✓			
2	User Interface I		✓	✓		
3	User Interface II		✓	✓		
4	Working with files and Shared Preferences		✓	✓	✓	
5	Working with Database		✓	✓	✓	✓
6	Web Service Integration				✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	-	1	-	-	-	-	-	-	-	1	3
CO2	3	-	-	-	-	-	2	-	-	-	-	-	2	2
CO3	-	3	3	-	-	-	-	-	2	1	-	-	3	2
CO4	-	3	3	2	2	-	1	-	-	2	2	3	3	3
CO5	-	-	3	3	3	2	-	-	-	1	3	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS324 : Game Development – II
(Marks 200)

Contact Hours: 06

Pre-requisite: MS221- Game Development - I

Methodology & Pedagogy: During theory lectures illustrations of 2D - 3D Graphics, animation and various concepts regarding Game Development will be discussed. Emphasis will be given on Unity Engine for Game Development. During Practical sessions, students will be developed Simple 2D Games, implement various features like animation, music and many more on that game objects, Handles Input and Output of data according to Game Object Interaction. Students will develop 2D Games using Unity Engine and also get some knowledge of 3D Game creation.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Hours	Numbers of
		Theory	Practical
1	Introduction to Unity Game Engine	04	36
2	Beginning with Unity	08	
3	Working with Unity UI System	06	
4	Scripting in Unity	06	
5	Navigation, Cursor Control and Functionality of Objects	05	
6	Work with Graphics and Audio Features	07	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit – I: Introduction to Unity Game Engine Hours: 04

Introduction of Unity Game Engine, History and Versions of Unity, Features of Unity, Asset Store of Unity, Tools and SDKs required for Mobile and Windows PC Game Development using Unity, Advantages of Unity Engine, and Basics of some c #concepts, Mono-behavior Life Cycle

Unit – II: Beginning with Unity Hours: 08

Creation of Project in Unity, Overview of Sprite System, Sprite Editor, Usage of Tags, Built in Game Object and Components, Rigid Bodies, Working with Classes – MonoDevelop, Terrain Management, Built in Game

Character, Controlling Game Character, Color and Texture ,Animation with Game Objects, Anchor and Pivot Points, Material and Map

Unit – III: Working with Unity UI System Hours: 06

Canvas, Text, Images, Button, Layouts, Toolbars, Menus, Creation and Transformation of Game Objects, World Co-ordinate system, Introduction to Scene, Camera: Camera Projections - Perspective (for 3D) and Orthographic (for 2D), Primitive Shapes, Snapping of Objects

Unit – IV: Scripting in Unity Hours: 06

Co-routine, Instance, Difference between C# Events and Unity Events, Conditions and States Delegates, Messaging, usage of Time (Class), Assigning Scripts to GameObjects, Passing data across scenes

Unit – V: Navigation, Cursor Control and Functionality of Objects Hours: 05

Navigate Objects, Collision, Unity GUI Cursor, Object to Object Communication, Destroy game object

Unit – VI: Work with Graphics and Audio Features Hours: 07

Introduction to 3D Objects, FBX - Importing 3D models in Unity Scenes, Shadow, Lights, Adding Music and Sounds, Controlling Sound and Music Volume, Colliders, Physics, Adding new Assets, Shaders, Loading external resource files: using Unity Default Resources, by downloading files from the Internet, Saving screenshots from the game

Core Books:

1. Greg Lukosek , “Learning C# by Developing Games with Unity 5.x”, 2nd Edition, Packt Publishing, 2016
2. Ashley Godbold, Simon Jackson, “Mastering Unity 2D Game Development”, 2nd Edition, Packt Publishing, 2016
3. Joe Hocking , “Unity in Action: Multiplatform game development in C#”, 2nd Edition, Manning Publications, 2018
4. Sue Blackman, “Beginning 3D Game Development with Unity 4: All-in-one, multi-platform game development”, 2nd Edition, Apress , 2013

Reference Books:

1. Ben Tristem, Mike Geig, “Unity Game Development in 24 Hours- Sams Teach Yourself” ,2nd Edition, Paperback, December 19, 2015
2. Michelle Menard, “Game Development with Unity”, Course Technology, 2013
3. Matt Smith, Chico Queiroz, “Unity 5.x Cookbook”, Packt Publishing, October 5, 2015
4. Francesco Sapio, “Unity UI Cookbook”, Packt Publishing, 2015
5. P Patrick Felicia, “Unity 5 from Zero to Proficiency: A step-by-step guide to creating your first game”, CreateSpace Independent Publishing Platform, February 25, 2016
6. Matt Smith, “Unity 2018 Cookbook”, 3rd Edition, Packt Publishing, August 31, 2018
7. Alan Thorn, “Mastering Unity Scripting”, Packt Publishing, January 29, 2015

Web References:

1. <https://unity3d.com/learn/tutorials/topics/developer-advice/how-start-your-game-development> [Game Tutorial]
2. <https://www.studytonight.com/game-development-in-2D/> [Game Tutorial]
3. <https://msdn.microsoft.com/en-us/magazine/dn759441.aspx> [Game Tutorial]

4. <https://www.youtube.com/playlist?list=PLPV2Kylb3jR42oVBU6K2DIL6Y22Ry9J1c> [Video Tutorial]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Be having basic knowledge of Unity Engine
- C02 : Be able to create simple games
- C03 : Be familiar with various UI and UI Event Handling
- C04 : Be able to implement inter communication between Game Objects.
- C05 : Be able to perform animation, apply different graphics features in the Game.
- C06 : Be able to create games with advance feature such as sound, music and many more.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Be able to work with Unity Engine	✓	✓	✓	✓	✓
2	Be able to create simple games	✓	✓		✓	✓
3	Be familiar with various UI and UI Event Handling		✓	✓	✓	
4	Be able to implement inter communication between Game Objects.		✓	✓	✓	✓
5	Be able to perform animation, apply different graphics features in Game.	✓	✓		✓	✓
6	Be able to create games with advance feature such as sound, music and many more.					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	2	-	1	-	-	-	1	2	3	-
CO2	2	1	3	2	3	-	1	-	-	-	-	1	2	-
CO3	-	2	2	1	-	-	1	-	-	-	-	-	1	-
CO4	1	3	3	2	-	-	2	-	-	-	-	1	2	-
CO5	-	-	1	-	1	-	1	2	-	-	1	-	2	-
CO6	-	1	-	3	2	-	-	-	-	-	-	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS325 : Data Analytics
(200 Marks)

Contact Hours: 06

Pre-requisite: MS117 : Database Technologies

Methodology & Pedagogy : During theory lectures the emphasis will be given on the basics of data analytics and related tools and techniques. Students will be introduced to the concepts of data science, exploratory data analysis, supervised and unsupervised learning methods. Applications as well as research trends and future direction of data analytics will be discussed with the length. During the practical sessions, students will be introduced to tools of data analytics such as R and Weka. Students will be given appropriate case studies of data analytics to get the real time exposure of data analytics.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Introduction of Data Analytics	06	36
2	Exploratory Data Analysis and the Data Science Process	06	
3	Supervised Learning Methods	07	
4	Unsupervised Learning Methods	07	
5	Applied Data Science	05	
6	Recent trends and future directions	05	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit – I Introduction of Data Analytics Hours: 06

What is Data Science? ,Big Data and Data Science hype ,Why now? , Datafication , Current landscape of perspectives , Skill sets needed

Unit – II Exploratory Data Analysis and the Data Science Process Hours: 06

Philosophy of EDA - The Data Science Process ,Statistical Inference , Populations and samples , Statistical modeling, probability distributions, fitting a model

Introduction to R ,Basic tools (plots, graphs and summary statistics) of EDA

Unit –III: Supervised Learning Methods Hours: 07

Supervised methods: Linear Regression ,Classification Trees, Random Forest, Neural Networks: Different

Models like single and multi-layer perceptron, back propagation, Application

Unit –IV: Unsupervised Learning Methods Hours: 07

Clustering, Association Rule Mining, Dimensionality Reduction - Singular Value Decomposition - Principal Component Analysis

Unit – V: Applied Data Science Hours: 05

Time Series Analysis, Recommendation Systems, Mining Social-Network Graphs

Unit – VI: Recent trends and future directions Hours: 05

Data Visualization - Basic principles, ideas and tools for data visualization, Data Science and Ethical Issues - Discussions on privacy, security, ethics , Next-generation data scientists, Basics of Big Data analytics

Core Books:

1. Cathy O’Neil and Rachel Schutt: Doing Data Science, Straight Talk From The Frontline, O’Reilly. 2014.
2. Jure Leskovec, Anand Rajaraman, and Jeffrey David Ullman, Mining of Massive Datasets Cambridge University Press, 2nd Edition, New York, NY, USA, 2014.
3. Howard B. Demuth, Mark H. Beale, Orlando De Jess, and Martin T. Hagan, *Neural Network Design* , paperback USA, 2nd Edition, 2014.

Reference Books:

1. Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K. , Probability & statistics for engineers & scientists ,9th edition, Prentice Hall, 2012.
2. Haykin, S. S., Haykin, S. S., Haykin, S. S., & Haykin, S. S., Neural networks and learning machines Pearson, Volume 3, 2009.
3. Mohammed J. Zaki and Wagner Miera Jr, Data Mining and Analysis: Fundamental Concepts and Algorithms., Cambridge University Press. 2014.

Web References:

1. https://onlinecourses.nptel.ac.in/noc17_mg24/preview [Online Data Analytics Course]
2. <https://www.itl.nist.gov/div898/handbook/eda/section1/eda11.htm> [Exploratory Data Analysis Material]
3. https://datahoarder.io/Humble%20Bundle%20Books/Humble%20Book%20Bundle_%20Data%20Science%20presented%20by%20O_Reilly/doingdatascience.pdf [Data Science E-Book]
4. http://www.astro.caltech.edu/~george/aybi199/Donalek_Classif.pdf [Supervised and unsupervised methods tutorials]
5. <https://yourstory.com/2017/12/data-analytics-future-trends/> [Future Trends of Data Analytics]

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Course Outcomes: Upon successful completion of the course, the students will :

- C01 : Describe what Data Science and Data Analytics are and the skill sets needed to be a data scientist.
- C02 : Understand significance of exploratory data analysis in statistical and visualization aspects.
- C03 : Understand and apply data analytics techniques such as supervised,unsupervised and EDA.
- C04 : Understand the importance of analytics in applications and able to implement them using data science techniques.
- C05 : Able to understand the recent trends and future directions of data analytics.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction of Data Analytics	✓				
2	Exploratory Data Analysis and the Data Science Process		✓	✓		
3	Supervised Learning Methods			✓		
4	Unsupervised Learning Methods			✓		
5	Applied Data Science				✓	
6	Recent trends and future directions					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	2	-	-	-	-	-	2	-	-	-	-	-
CO2	-	1	3	2	-	-	-	-	-	-	-	-	-	-
CO3	-	3	-	2	-	1	1	-	-	-	-	-	-	-
CO4	-	-	-	3	-	-	2	1	-	-	-	-	-	-
CO5	3	1	2	-	1	-	1	-	1	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS326 : Ruby on Rail Architecture

(200 Marks)

Contact Hours: 06

Pre-requisite: None.

Methodology & Pedagogy: During theory sessions topics related to common technologies and techniques used in the developing of web-based applications will be covered with suitable examples.

During Practical sessions, students will be required to design and develop entire web sites using Ruby On Rails.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of hours	
		Theory	Practical
1	Getting started and basics of Ruby Language Literacy.	06	36
2	Working with standard types, Methods and Expression	05	
3	Working with Arrays, Hashes and Enumerable	06	
4	Working with Times and Dates, Exceptions, Catch and Throw	05	
5	Files, I/o and Databases	07	
6	OOP and Dynamic features in Ruby	07	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit – I: Getting started and basics of Ruby Language Literacy Hours: 06

What is Ruby; Ruby download and installation, An Introduction to Object Orientation, Basic Ruby Syntax and Semantics OOP in Ruby, Dynamic Aspects of Ruby.

Unit – II: Working with standard types, Methods and Expression Hours: 05

Numbers, Strings, Ranges, Regular Expressions, defining a method, Calling a method, Operator expressions, Miscellaneous expressions, Assignment, Conditional execution, Case expression, Loops, variable Scope, Loops and Blocks.

Unit – III: Working with Arrays, Hashes and Enumerable Hours: 06

Comparing arrays, Sorting array, selecting from an array by criteria, using multidimensional arrays, Finding, transforming, removing, Concatenating and reversing in arrays and hashes with key value pairs.

Searching and selecting, Counting and comparing, iterating, Extracting and converting in Enumerable

Unit – IV: Working with Times and Dates, Exceptions, Catch and Throw Hours: 05

Determining and finding time and dates, conversion, parsing and formatting of date and time. Exception class, Handling and Raising exceptions, Catch and Throw.

Unit – V: Files, I/o and Databases Hours: 07

What is an IO object, Opening and closing files, Reading and writing files, Connecting to external data stores like MySQL, ORM and SQLite.

Unit – VI: OOP and Dynamic features in Ruby Hours: 07

Revision of OPP concepts, Learning some advance techniques and dynamic features and Program introspection.

Core Books:

1. Hal Fulton, Andre Arko: The ruby way, Third edition, Addison Wesley
2. Dave Thomas, Chad Fowler, Andy Hunt: Programming Ruby -The Pragmatic Programmers' Guide, Second Edition, The Pragmatic Bookshelf
3. David A. Black: The Well-Grounded Rubyist, Manning Publications Co., Greenwich

Reference Books:

1. Peter Cooper: Beginning Ruby, Second Edition, Apress
2. Chris Pine: Learn to Program, Second Edition, The Pragmatic Bookshelf

Web References:

1. <https://www.railstutorial.org/book> [For additional reading of web development using Ruby On Rails]
2. <https://learnrubythehardway.org/book/> [For additional reading from references]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Understand the concept of Ruby on Rail with its basics terminologies.
- C02 : Get aware with data types, operators and controls statements available in Ruby on Rail.
- C03 : Learn various ways to store multiple values in single variable.
- C04 : Be able to handle exception.
- C05 : Be able to work with different types of Database from Ruby on Rail.
- C06: Be able to implement object oriented concepts to create real world application.

Course Outcomes Mapping:

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Unit No.	Unit Name	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1	Getting started and basics of Ruby Language Literacy	✓					
2	Working with standard types, Methods and Expression		✓				
3	Working with Arrays, Hashes and Enumerable			✓			
4	Working with Times and Dates, Exceptions, Catch and Throw				✓		
5	Files, I/o and Databases					✓	
6	OOP and Dynamic features in Ruby						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	2	2	2	1	-	-	-	-	-	-	-	3	3
CO2	3	2	2	3	2	2	-	3	-	-	-	-	2	3
CO3	-	-	3	3	2	-	-	-	-	-	-	-	2	3
CO4	3	2	2	-	2	2	-	3	-	-	-	-	3	3
CO5	2	-	-	-	2	-	3	3	-	-	-	-	3	2
CO6	-	2	2	2	1	-	-	-	-	-	-	-	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS327: Internetworking with TCP/IP
(Marks 200)

Contact Hours: 06

Pre- requisite: MS118 – Fundamentals of Computer Networks.

Methodology & Pedagogy: During theory sessions the students shall be introduced to various internet and intranet technologies and network services. As a case study an institute level network design, devices, components and security mechanism will be demonstrated. During practical sessions students will be trained to develop network based applications using available technologies.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Numbers of Hours	
		Theory	Practical
1	Addressing Techniques	06	36
2	Internet Protocol (TCP/IP)	06	
3	Routing	07	
4	UDP and Mobile IP	06	
5	Cellular Systems	05	
6	Internet Security	06	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total: 72

Detailed Syllabus:

Unit – I: Addressing Technique Hours: 06

Classful Internet Addresses, Mapping Internet Addresses (ARP), Determining An Internet Address (RARP)

Unit – II: Internet Protocols (TCP/IP) Hours: 06

Internet Protocol: Connectionless Datagram Delivery, Internet Protocol: Routing IP Datagrams, Internet Protocol: Error and control Messages, ICMP

Unit – III: Routing Hours: 07

Congestion, Router Discovery and solicitation, Classless and Subnet Address Extensions, Transparent routers, Proxy ARP and Subnet addressing, CIDR, Subnet routing. Static and Dynamic routing, Difference between static and dynamic routing. Introduction to dynamic routing algorithms.

Unit – IV: UDP and Mobile IP Hours: 06

User Datagram Protocol, Reliable Stream Transport Service, Sliding Window, TCP, Karn's Algo, Congestion, RED Silly window syndrome, starting and closing TCP connection, Mobile IP

Unit – V: Cellular Systems Hours: 05

Cellular system overview - Cellular system organization, Frequency Reuse, Increasing Capacity, Operation of cellular system, steps in an Mobile switching center(MSC) controlled call between mobile users, Mobile Radio Propagation effect, Additional Function in MSC Controlled Call, Handoff Performance metrics, Handoff Strategies Used to Determine Instant of Handoff, Power control, Traffic Engineering.

Unit – VI: Internet Security Hours: 06

Electronic mail, TCP/IP, SMTP, MIME, Applications: World Wide Web Internet Security and Firewall Design, Internet Security, Firewalls

Core Books:

1. Duglas E Comer: Internetworking with TCP/IP Volume – I, Fourth Edition, PHI.
2. Andrew S. Tanenbaum, David Wetherall: Computer Networks, PHI 3. Ajit Pal: Data Communications and Computer Networks, PHI

Reference Books:

1. Behrouz A. Forouzan: TCP / IP Protocol Suite, Third Edition
2. Natalia Olifer, Victor Olifer: Computer Networks: Principles, Technologies and Protocols for Network Design, Willy
3. Debra Littlejohn Shinder: Computer Networking Essentials: An essential guide to understanding networking theory, implementation and interoperability, Cisco Press.

Web References:

1. <http://www.firewall.cx> [Basics of Network Protocols]
2. <http://nptel.ac.in> [Video Lectures]
3. <https://www.cs.purdue.edu/homes/comer/netbooks.html> [Teaching Material]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Better understanding of classful and classless addressing scheme.
- C02 : Be familiar with TCP/IP reference model.
- C03 : To understand different routing protocols.
- C04 : To familiar themselves with mobility in communication.
- C05 : To familiar themselves with cellular systems communication.
- C06: To familiar themselves with network security.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		C01	C02	C03	C04	C05	C06
1	Addressing Techniques	✓					
2	Internet Protocol (TCP/IP)		✓				
3	Routing			✓			
4	UDP and Mobile IP				✓		
5	Cellular Systems					✓	
6	Internet Security						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	2	-	1	-	-	-	1	2	3	-
CO2	2	1	3	2	3	-	1	-	-	-	-	1	2	-
CO3	-	2	2	1	-	-	1	-	-	-	-	-	1	-
CO4	1	3	3	2	-	-	2	-	-	-	-	1	2	-
CO5	-	-	1	-	1	-	1	2	-	-	1	-	2	-
CO6	-	1	-	3	2	-	-	-	-	-	-	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

TEACHING SCHEME & DETAILED SYLLABUS

FOR

**M.Sc. (IT) PROGRAMME
(4th SEMESTER)**

EFFECTIVE FROM ACADEMIC YEAR 2021-22

Teaching and Examination Scheme (M.Sc(IT) Programme)
Effective from Year 2021-22

SEMESTER-IV

Course Code	Course Title	Teaching Scheme				Internal	End Semester Examination		Total
		Contact Hours			Credit	Continuous Evaluation	Report	Presentation & Viva	
		Inst.	Industry	Total					
MS402	Dissertation/Project Work	2	28	30	30	200	200	400	800

MS402 : Dissertation/Project Work

Guidelines to Undertake the Final Semester M.Sc.(IT) Project Work

Project/Dissertation Work:

Students of M.Sc.IT have to do the Project/Dissertation work in an Industrial/ Research Organization of computer field. Project/Dissertation work shall be carried out under the supervision of a qualified teacher in the Department as well as an expert from organization. Students have to meet the institute supervisor periodically and to attend the project/dissertation review meetings for evaluating the progress. The Dissertation/Project work shall be pursued for a minimum of 16 weeks during the semester.

1.1 MORI Principle to choose the topic:

Final semester dissertation/project work is important and challenging element of MCA study. Selection of topic of project/dissertation work is very crucial and important aspect of that. Following are important principles that will help students to select the appropriate topic of project/dissertation work.

- **Manageable:** your dissertation topic must be sufficiently focused so that it is possible for you to do the topic justice within the available time (one semester). You may have a real interest in, say, *'the impact of technology computers on Indian Economic Growth since Independence'*, but you certainly won't be able to cover this topic in any detail in the space of one semester (or around 5-7 thousand words)!
- **Original:** this relates to the above point, since a topic that is focused and manageable is more likely to be one that has not been written about too extensively, thus leaving room for your original contribution. Ideally you will find an interesting and well-chosen topic which will impress those marking your work.
- **Relevant:** your project should clearly be relevant to some aspect of your studies, but it might also be relevant to your plans for, say, postgraduate study or a career. The dissertation may also be relevant in the sense that it plays to some of your established strengths, such as a particular course module or topic that you have enjoyed studying and in which you have previously done well.
- **Interesting:** you are obviously more likely to enjoy and be successful in your dissertation if it is of real interest to you and to those marking your work. Ask yourself if you are sufficiently committed to your idea to be able to give it your best throughout the duration of your project. You should also ascertain whether your supervisor finds the idea interesting during your initial discussions with her or him.

1.2 Evaluation Of Project/Dissertation Work:

The dissertation / project report shall normally be written in English in the specified format and shall be characterized by significant contribution to knowledge in the field. Normally two copies of the report are to be submitted for evaluation. The Project/Dissertation report prepared according to approved guidelines and duly signed by the supervisor(s) and the Head of the Department shall be submitted to the Head of the Institution. The evaluation scheme of Project/Dissertation is as under:

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Course	Course Title	Teaching Scheme				Internal	End Semester Examination		Total
		Contact Hours			Credit	Continuous Evaluation	Report	Presentation & Viva	
		Inst.	Industry	Total					
MS402	Dissertation/Project work	2	28	30	30	200	200	400	800

The internal evaluation of project/dissertation is done based on progress reports and internal presentations.

The final evaluation of the project/dissertation will be based on the project report submitted and a VivaVoce Examination by a Board of Examiners.

If a candidate fails to submit the project report on or before the specified deadline, he/she is deemed to have failed in the Project /Dissertation Work and shall re-enroll for the same in a subsequent semester. If a candidate fails in the viva-voce examinations of Project/ Dissertation work, he/she shall resubmit the project report within specified duration decided by university. The resubmitted project will be evaluated during the subsequent academic session. A copy of the approved project report after the successful completion of viva examinations shall be kept in the library of the college / institution.

Web References:

1. <http://www.microtoolsinc.com/Howsrs.php> [For Software Requirement Specification Guidelines]
2. <http://techwhirl.com/writing-software-requirements-specifications/> [For goals of SRS]
3. <https://kepler-project.org/developers/reference/software-development-guidelines> [For Software Development Guidelines]

Course Outcomes:

CO1 : Student will understand the implementation of concepts of SDLC and Software Engineering.

CO2 :The programming concepts they learn during their academics, it will be converted in to the actual implementations.

CO3: Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.

CO4: Students must understand the difference between a program and professional application/product/software.

CO5: Students will learn different categories of applications like Desktop application, Web applications, etc.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO2	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO3	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO4	3	3	3	2	3	3	3	3	3	2	2	2	3	2
CO5	3	3	3	2	3	3	3	3	3	2	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”